

FETCH: INVENTORY MANAGEMENT SYSTEM WITH PREDICTIVE ANALYTICS FOR RISHA PET SUPPLIES

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By

JUGAN, JOYLINDA P.
REUSORA, JUAN CARLO M.

Chapter 1

The Problem and a Review of Related Literature

Introduction

The pet care sector around the world is doing really well now. Grand View Research said in 2023 that the pet care industry will be worth $350 billion by 2027. The reasons for this are simple. People think about their pets in a way now. Pets are not just animals, they are part of the family. Because of this people are spending money on good food, special things and good health care for their pets. The pet care industry is growing because people consider their pets to be family members. People want the best for their pets so they buy them food and quality healthcare items.

This seems like a chance for people who own pet stores but it is actually very hard for small business owners like the ones at Risha Pet Supplies. Think about what it takes to run a store. You have to deal with a lot of things like food, grooming supplies, toys and accessories. These are a few examples of the many things you have to manage.

How do small pet stores handle all this stuff? They do it by hand. This means they have to count how many things they have in stock, write down numbers in spreadsheets and decide when to order things based on what they remember and how they feel. Managing a store, like Risha Pet Supplies, is not easy because you have to keep track of so many pet supplies.

The problems with this approach are very clear. Chen and Lee found in 2022 that these old ways of doing things cause some problems. For example, popular items sell out. That means lost sales and unhappy customers. At the time other products just sit on

shelves, which means the company is wasting money and space. Bad decisions about what to order can also hurt relationships with suppliers, like Chen and Lee talked about. Small businesses feel these problems a lot because they have to be very careful with money. They do not have a lot of money to spare. This means that if they make a mistake with the things they are selling it can be very expensive. Small businesses have to watch every penny they spend because they do not make a lot of profit. If they make a mistake with the inventory it can be very bad for the business. Small businesses really need to get the inventory just right so they do not lose money.

Intelligence and machine learning are really helping us with some big problems we have been dealing with for a long time. With analytics we can find sales patterns that people would normally miss. For example sales might go up a bit on weekends or they might change with the seasons or some products might be related to each other in ways we do not expect. Kumar et al. (2023) Talk about how intelligence and machine learning are helping businesses change the way they manage their inventory. By waiting for products to sell out and then trying to reorder them stores can now use artificial intelligence and machine learning to forecast what they will need and place their orders ahead of time. This means stores can stay on top of intelligence and machine learning to manage their inventory in a better way. The results can be really big: Waller and Fawcett found out in 2021 that companies that use Artificial Intelligence inventory systems can cut their costs by 20 to 50 percent. At the time these Artificial Intelligence inventory systems helped companies make their products available to people when they needed them.

This study is about creating a system to manage stock for Risha Pet Supplies. The system uses a combination of two methods, Random Forest and Gradient Boosting to make predictions about what people will buy. Risha Pet Supplies will use this system to predict what they will sell, figure out when they need to order and send messages to their suppliers. This is helpful for a store like Risha Pet Supplies because they have to deal with things like people buying more during certain times of the year, products that go bad after a while and pet owners who really like certain brands. The system for Risha Pet Supplies will make it easier for them to manage all of this.

This research fills an important gap in the existing literature, which tends to focus primarily on large retail operations rather than small pet supply businesses. Beyond that, it demonstrates how small retailers can leverage advanced technology to compete more effectively against large chains and e-commerce giants, showing that sophisticated inventory management tools are genuinely accessible beyond large corporations with extensive IT resources.

Review of Related Studies and Literature

This section brings together past research and existing systems that are relevant to the development of an intelligent inventory management system for Risha Pet Supplies. It focuses on practical and tested solutions, showing how inventory management has gradually evolved from manual tracking and basic formulas to more advanced, data-driven approaches. The discussion also looks at how technologies such as machine learning and predictive analytics are being used in retail to improve forecasting and stock control. Special attention is given to small and medium enterprises, since these businesses

often operate with limited resources and face challenges that are very different from large companies. Finally, the review considers the unique inventory needs of pet retailers, ensuring that the proposed system is aligned with the real-day-to-day operations of a pet supply store. It also reviews User Acceptance Testing (UAT) and the Computer System Usability Questionnaire (CSUQ), the tools that will be used to test and evaluate the system.

Inventory Management in Retail Operations. The conventional approach using Economic Order Quantity (EOQ) relies on mathematical models in purchase optimization but lacks in a retail business with a constantly fluctuating demand. Chen and Lee (2022) report changes in purchase behavior with the effects of seasons, promotions, and consumer behavior, posing a problem, especially in smaller businesses with mounting analytical capacity similar to conventional large enterprises. The problem of the aforementioned gap in analytical capacity in an SME with the use of predictive analysis beyond the conventional EOQ approach in purchase optimization in Risha Pet Supplies is addressed by the proposed FETCH.

Machine Learning Applications in Inventory Management. Machine learning has brought forth an unprecedented paradigm shift in accurate and quantifiable results in retail demand forecasting. A study conducted by Shirkhorshidi et al. in 2022, including analysis using 50 stores, validated that an ensemble model combining Random Forest and Gradient Boosting reached 94.3% accuracy, compared to 78.6% using conventional ARIMA, and led to a reduction in stockout products by 42% and overstock products by 35%. Li et al. in 2024 illustrated that Gradient Boosting has been successfully utilized to

uncover patterns in seasonality, which are essential in pet stores, where merchandise changes rapidly season to season. FETCH will adopt the same combination of Random Forest and Gradient Boosting and will measure the accuracy of its own forecasts on the sales of Risha Pet Supplies, so that its results can be compared with this benchmark.

Ensemble Machine Learning Algorithms: Random Forest and Gradient Boosting. Ensemble algorithms combine multiple models together to achieve better accuracy. Hastie et al. (2020) describe how the Random Forest creates multiple decision trees from random subsets of data and then pools the predictions to reduce overfitting, giving feature importance rankings. Gradient Boosting creates models in sequence to correct errors of the previous model by iteration. FETCH leverages both algorithms' complementary strengths: Random Forest provides robust baseline predictions resistant to noise, while Gradient Boosting refines predictions by learning residual patterns, excelling at seasonal pattern identification critical for pet retail.

Predictive Analytics and AI-Driven Forecasting in Retail. The incorporation of AI approaches led to enhanced accuracy of forecasting by 23% and enhanced supply chain optimization by 28% relative to conventional approaches as revealed by Avula (2024). Moreover, it was also revealed by McKinsey in 2021 that those organizations which applied AI-based forecasting experienced a reduction of 20-50% in their supply chain error rates, thereby recording an improvement in efficiency of up to 65%. The FETCH system uses these strategies in handling historical transaction data, seasonality data, and promotion data. Thus, the automatic reorder feature of FETCH will find the products that have reached their reorder level, work out how many of each to order from the AI

forecast, and create and email a purchase order to each supplier once the owner confirms, avoiding delays caused by human error.

***Automated Inventory Systems with IoT Integration*.** Ibrahim et al. mentioned in 2024 that companies using the ERP/IoT-based system have increased their inventory turnover by a reported 45%, reduced stockouts by a reported 50%, and achieved a reported 95% accuracy of inventory compared to the human-controlled system's accuracy of 75% to 85%. Similarly, the FETCH system is designed on the same lines, aiming for automation with the help of barcode scanning and auto-reorder. The application of the barcode scanning system enables the fast identification of the items under the auto-updated system. The auto-indication of purchase order generation to the supplier can result in a reported 50% reduction in stockouts and a reported 45% improvement in inventory turnover.

Agile Software Development Life Cycle (SDLC). It is documented that Agile Software Development Life Cycle focuses on iterative development with an emphasis on flexibility and delivering the product continuously through short sprint cycles. Rigby et al. (2020) showed Agile's focus on iterative development in terms of flexibility with successful rates and better quality. The FETCH model applies to the Agile Software Development Life Cycle by continuously delivering functional features with stakeholder feedback.

Technology Acceptance and User Adoption in Retail Systems. The perceived usefulness and ease of use significantly influenced system acceptance, while another project that involved end-users saw the acceptance of a system increase to 89% from 56%,

as reported by Ullah et al. (2024). The FETCH solution includes "Agile SDLC to ensure the incorporation of feedback, followed by User Acceptance Testing (UAT), where the functional compliance of the system is tested to the fullest. The Computer System Usability Questionnaire (CSUQ) measures perceived usability under System Usefulness, Information Quality, Interface Quality and Overall Satisfaction."

User Acceptance Testing (UAT). User Acceptance Testing checks whether a system meets the needs of the people who will use it in their actual work. Pillai and Hemamalini (2022) showed that a hybrid UAT procedure, started early in development and repeated through the stages of the life cycle, detects defects earlier, makes debugging more efficient and lowers the risk of failure after deployment, compared with testing only at the end as in the Waterfall approach. Their study also found that UAT does more than show that a system works: it shows that it works for the people who will use it, which builds their trust in the system. FETCH applies this through two levels of UAT. In unit testing, technical experts test each module with completeness, format, range, consistency and database checks. In system testing, the end users of Risha Pet Supplies (the administrator, the cashier and the suppliers) carry out their own tasks screen by screen, from signing in to logging out.

Computer System Usability Questionnaire (CSUQ). The CSUQ is a standardized questionnaire that measures how usable a computer system is, as perceived by its users. Lewis (1995) developed it at IBM and reported coefficient alphas of 0.89 and above, showing that it is highly reliable. In a review of the questionnaires used to evaluate satisfaction, usability and acceptance in mobile health studies, Hajesmaeel-Gohari et al.

(2022) found the CSUQ and its counterpart, the Post-Study System Usability Questionnaire (PSSUQ), among the most used instruments. FETCH uses the CSUQ to measure System Usefulness, Information Quality, Interface Quality and Overall Satisfaction from each type of user.

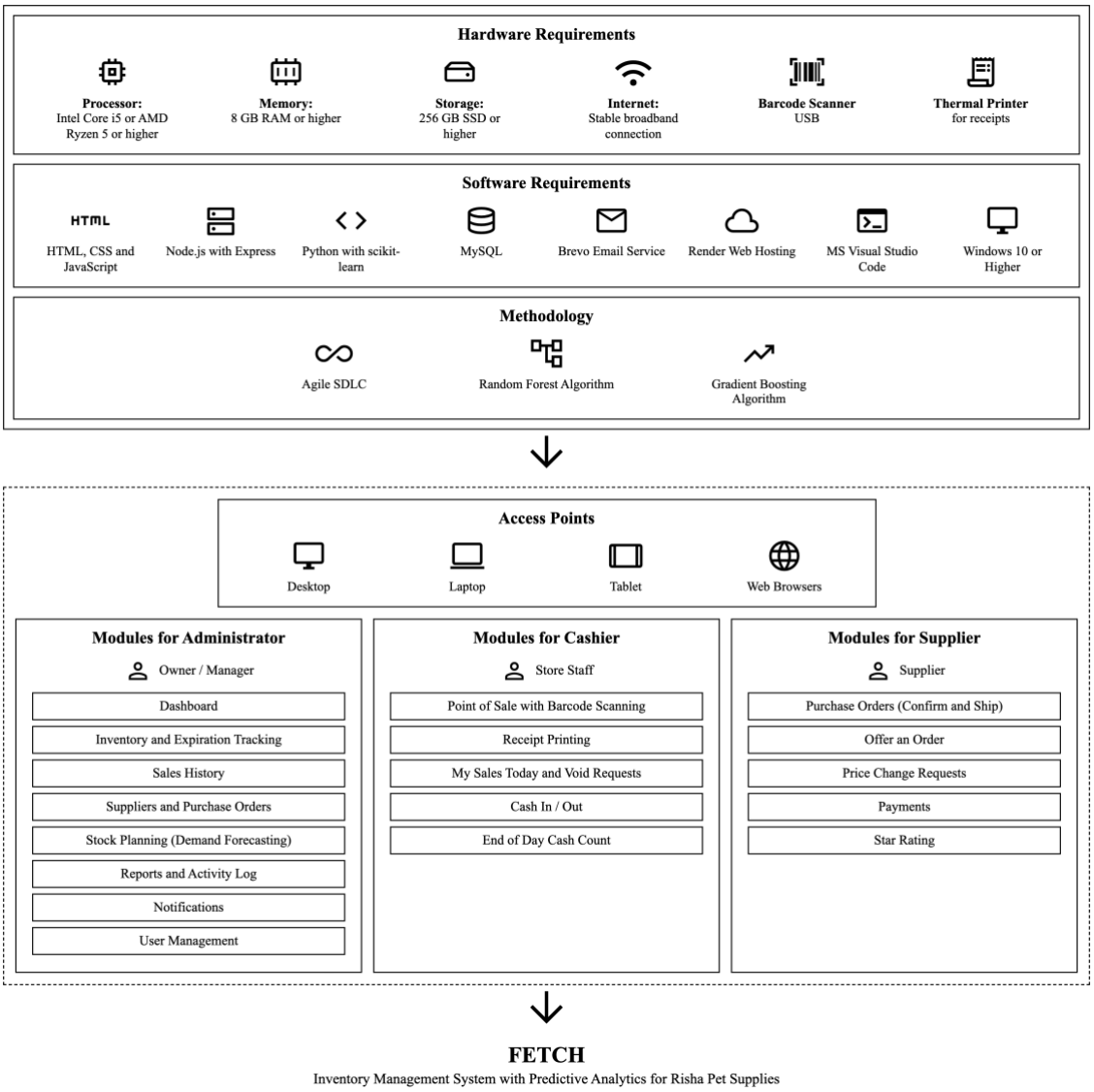
Pet Retail Industry: Characteristics, Challenges, and Opportunities. The retail industry for pets also comes with its own set of challenges due to the diverse nature of the products. According to Leafio (2024), “The basic product groups are characterized by a few hundred SKUs with different shelf life arrangements." Seasonal demand patterns, including higher sales of flea protection items in the Summer and Q4 sales of holiday items, have been observed, as reported by RELEX Solutions (2023). Industry-related challenges offer a need for the development of FETCH. Pattern Learning deals with the Seasonal Variability issue, Barcode Scanning is capable of handling many SKUs along with the expiration date, and Automatic Reorder tackles the issue of Overstocking with less likelihood of Stockout.

Conceptual Diagram

This section presents the schematic diagram as a visual representation of the FETCH system architecture. It shows how the hardware, software, methodology, access points and the modules of each user are designed to work together to achieve the objectives of the study.

Figure 1

Schematic Diagram of FETCH



Illustrated in Figure 1 is the architecture of FETCH, which is built to replace the manual inventory tracking of Risha Pet Supplies. The top part shows the hardware and software the system needs and the methodology used to build it: the Agile SDLC, and the Random Forest and Gradient Boosting algorithms that forecast demand. FETCH is a web-

based system, so its users open it through a web browser on a desktop, laptop or tablet. Each user sees only the modules of their role. The administrator, who is the owner or manager, manages the inventory, sales, suppliers, purchase orders, users and reports, and uses Stock Planning to see the demand forecast and what to reorder. The cashier, who is the store staff, uses the Point of Sale to scan products, complete sales, print receipts and count the cash at the end of the day. The supplier uses the Supplier Portal to confirm and ship purchase orders, offer orders, request price changes, and see its payments and star rating. Every sale, delivery and stock adjustment updates the same database, so all three users see the same figures.

Hardware Requirements. The computer used to develop and run FETCH will require an Intel Core i5 or AMD Ryzen 5 processor or higher to handle the Random Forest and Gradient Boosting models, at least 8 GB of RAM so that the point of sale, the inventory pages and the forecast can run at the same time, and a 256 GB SSD or higher for fast access to the sales records. A stable internet connection will be needed, since FETCH is a web-based system that is opened through a web browser and sends emails to suppliers. A barcode scanner will be used to identify products at the point of sale, and a thermal printer will print the receipts.

Software Requirements. FETCH will be developed on Windows 10 or 11 using Microsoft Visual Studio Code. The front end will use HTML, CSS and JavaScript to give the employees of the pet shop a clean and easy interface. The back end will use Node.js with the Express framework, and the demand forecasting will use Python with the scikit-learn library, which provides the Random Forest and Gradient Boosting algorithms.

MySQL will be the main database. FETCH will be hosted online on Render so that it can be opened from any web browser, and it will send emails to suppliers and users through the Brevo email service.

Software Development Lifecycle Model. The software development project will adopt the Agile Software Development Model. Through the Agile software development approach using sprints, the researchers will develop and test specific features of the system such as the AI forecasting engine and the stock alert system feature incrementally. Through the Agile approach, the system will get continuous feedback from the management of Risha Pet Supplies so that the system will be fine-tuned based on actual experience before the final deployment of the system.

Access Points. FETCH will be web-based and can be opened through a web browser on a desktop, laptop or tablet. To keep the data secure, each account will have one of three roles, and each role will see only its own pages. Through the Administrator Access Point, the store owner or manager will monitor all operations: the inventory, sales, suppliers, purchase orders, reports, the demand forecast and the user accounts. Through the Cashier Access Point, the store staff will go straight to the Point of Sale, a simple screen for scanning products, completing sales, recording cash taken in and out, and counting the cash at the end of the day. Through the Supplier Access Point, each supplier will see only its own purchase orders and products, where it can confirm and ship orders, offer an order, request a price change, and see its payments and star rating.

Inventory Tracking Module. This particular module will enable automation in the tracking of pet products. This will be achieved by recording crucial details such as Stock

Keeping Units, Expiration dates, and Category Types. This particular module will take control away from the store's logbooks, which are currently being used, and will enable real-time tracking for all available products. This will be achieved through automatic updates when shipments arrive at the store, as well as when sales are made.

Predictive Analytics & AI Module. It is set to function as the backbone of the system. It will use two machine learning algorithms, Random Forest and Gradient Boosting, trained on the historical sales of the store, and combine their predictions to forecast how many of each product will sell in the next 30 days. It will learn patterns such as more of a particular brand of dog food being bought on weekends or holidays, and use the forecast to tell the business owner which products to reorder, how many and by when.

Point of Sale (POS) and Transaction Module. The cashier will scan a product's barcode with a barcode scanner or choose it on the screen, and the system will complete the sale, print the receipt and subtract the items from the stock at once. The system will automatically stamp the date, the time and the cashier on each transaction, which forms the raw data for the reports and the demand forecast.

Reporting & Notifying Module. The system will help the user keep the right amount of stock through visual reports and notifications. It will send "Low Stock" notifications once a product reaches its reorder level and will email purchase orders to the suppliers, giving the owner a chance to restock in time. The module will also give the management of Risha Pet Supplies detailed reports on sales trends to guide their decisions.

Supplier Management Module. The system will keep the record of each supplier and of the purchase orders sent to them. When the owner creates purchase orders from the reorder list, each supplier will receive its order by email and in the Supplier Portal, where it can confirm the order, set the delivery date and mark it as shipped. Receiving an order will add the delivered items to the stock. Each supplier will also be given a star rating based on its delivery record, such as whether its orders arrive by the promised date.

Objectives of the Study

This proposed research work aims to develop an intelligent inventory system for Risha Pet Supplies to revamp conventional inventory management systems through a combined platform involving machine learning algorithm-driven demand forecasting, automated inventory management, and supplier communication.

What the study will do:

1. To develop FETCH: an Inventory Management System with Predictive Analytics for Risha Pet Supplies that implements ensemble machine learning algorithms (Random Forest and Gradient Boosting) to predict future demand based on historical sales patterns and seasonal trends, integrates barcode scanning technology for rapid product identification and automated inventory updates, applies an automated reordering mechanism that generates purchase orders for products at or below their reorder points, maintains real-time tracking of stock quantities and expiration dates, and provides a supplier portal to facilitate communication and coordination with suppliers.

2. To test the system's functionality and acceptance through User Acceptance Testing (UAT), which will include:

    2.1. Unit Testing by Technical Experts

        2.1.1. Completeness Check

        2.1.2. Format Check

        2.1.3. Range Check

        2.1.4. Consistency Check

        2.1.5. Database Check

    2.2. System Testing by End Users

        2.2.1. Administrator (Store Owner or Manager)

        2.2.2. Cashier (Store Staff)

        2.2.3. Supplier

3. To evaluate the usability of the system using the Computer System Usability Questionnaire (CSUQ) in terms of:

    3.1. System Usefulness (SYSUSE)

    3.2. Information Quality (INFOQUAL)

    3.3. Interface Quality (INTERQUAL)

    3.4. Overall Satisfaction

Significance of the Study

The emergence of FETCH has great worth in enhancing the efficiency, profitability, and overall management procedure associated with the pet products industry. As an all-inclusive system that comprises stock management, predictive analysis, point-

of-sale system processing, and connectivity to supply chain networks, FETCH provides solutions to some of the problems associated with small to medium-sized stores, which may include stock inaccuracies, overstocking, stockout situations, manual processing errors, and poor supplier coordination. Some of the important contributions associated with this study are discussed below:

Risha Pet Supplies. The system enables all aspects of inventory management and analysis, as well as point-of-sale functionality. The desire is to ensure there is neither an excess nor a deficit and, accordingly, makes the correct predictions for stock and adjusts it according to the patterns and cycles. This would enable better management of cash flow and hence reduce the cost of carrying. Manage supplier communication effectively, order products for re-ordering to save time and reduce human error, and adjust your inventory levels after each sale in real-time to ensure you are as up-to-date as possible concerning your levels.

Suppliers. The supply chain management part of the platform enables a whole lot of coordination between Risha Pet Supplies and the suppliers. The process of ordering is also automated with real-time inventory visibility. The suppliers also have accurate forecasts of demand made possible by predictive analytics so as to produce and deliver in an optimized way. This actually makes it possible for the company to build better business relationships based on better communication or information flow.

Customers. Customers will get the benefits of increased product availability, as the predictive analytical tool is used to ensure that popular products are always available. They can also complete transactions faster and more accurately, as the POS is integrated

into the system. They are also assured that they will not get or purchase expired goods, especially pet foods and medication, which will improve customer satisfaction and safety. Customers will have a better experience while shopping, as they will get what they want through a much more effective and efficient means.

Business Owners and Managers in the Pet Supply Industry. This research work is very helpful to pet supplies retail business as it offers them a model for modernizing the way they handle their inventory. The model put forward demonstrates how well the use of predictive analysis can be applied in a pet supplies retail business environment, where the business encounters various difficulties such as a large variety of supplies, items, or goods. The study gives them awareness of the benefits a combined system, which handles inventory, sales, and vendors, can offer.

Future Researchers. This study can greatly help a researcher who aspires to have an inventory management system with predictive analytics and business process automation capabilities. Methods and design strategies for this study can shed light on future studies which can concentrate on more aspects of optimizing this system, more on forecasting, and also on how this automatic management of inventory can sustain business performance on its own and more. This study can be a stepping stone for other studies related to other advanced technologies as well, such as artificial intelligence, machine learning, and other retail business aspects related to pet supplies.

Scope and Delimitation of the Study

The proposed research works on the development of an Intelligent Inventory Management System with Predictive Analysis capabilities designed specifically for Risha

Pet Supplies. The system comprises inventory management with point of sales (POS) transactions, demand forecasting based on machine learning techniques (Random Forest & Gradient Boosting), automatic reordering of products, handling of suppliers, as well as expiration dates of pet supplies categorized as food items, accessories, healthcare products, and pet grooming products intended mostly for dogs & cats. The system will be developed using Node.js with the Express framework as the back end, Python with the scikit-learn library for the machine learning models, MySQL as the database, and HTML, CSS and JavaScript as the front end. It will be a web-based system with three types of users: the administrator, the cashier and the supplier.

The scope is narrowed down to Risha Pet Supplies and will not include other selling businesses and industries. The system will not include multi-branch synchronization, e-commerce, mobile apps, CRM, loyalty programs, and full-blown accounting. Predictive analytics will just include historical sales forecasting and will not include market trends, competitor analysis, economic, and social trends among others. Predictive analytics will be trained and developed using Risha Pet Supplies' historical sales and will not include any form of benchmarking from other businesses. The system will also not include multi-currency and different regions, which include different currencies from other nations. Additionally, it will not be multi-lingual and will be developed solely to cater to local operations in the Philippines. The system will also not include RFID, IoT, and other automatic dispensary systems and will include just barcode scanners and receipt printing. Performance analysis will be conducted just during implementation and will no longer include extended periods to test return on investment.

Security will include user log-in with role-based access, so that each user sees only the pages of their role, and email confirmation of new accounts; it will not include other advanced security features such as blockchain, biometrics and penetration testing.

Definition of terms

To ensure clarity and consistency throughout this study, the following section presents the key terms used in the research. Since the system developed for Risha Pet Supplies involves a combination of technical processes, data-driven methods, and inventory management concepts, each term is defined based on how it is applied within the study or its standard dictionary meaning. These definitions aim to provide readers with a clear understanding of the technical language used, allowing them to follow the discussions, analyses, and findings more effectively.

Automated Reordering. A process that is operated or controlled by automation for the purpose of ordering again (Merriam-Webster, n.d.-y; Merriam-Webster, n.d.-n).

Barcode Scanning. The process of reading a code consisting of printed and variously patterned bars and spaces into computer memory (Merriam-Webster, n.d.-b).

Computer System Usability Questionnaire (CSUQ). In this study, the CSUQ refers to the standardized questionnaire answered by the users of FETCH to rate its System Usefulness, Information Quality, Interface Quality and Overall Satisfaction (see Appendix A).

Database Management System. A usually large collection of data organized especially for rapid search and retrieval combined with a set of procedures and techniques

for managing an organization or a particular function (Merriam-Webster, n.d.-ad; Oxford Reference, n.d.-a).

Demand Forecasting. The process of calculating or predicting the quantity of a commodity or service wanted at a specified price and time, usually as a result of study and analysis of available pertinent data (Merriam-Webster, n.d.-c; Merriam-Webster, n.d.-d).

Demand Pattern. A reliable sample of traits, acts, tendencies, or other observable characteristics regarding the quantity of a commodity or service wanted at a specified price and time (Merriam-Webster, n.d.-ag; Merriam-Webster, n.d.-d).

Ensemble Learning. In this study, ensemble learning refers to the combination of multiple machine learning algorithms (Random Forest and Gradient Boosting) working together to produce more accurate inventory forecasts than any single algorithm could achieve independently for Risha Pet Supplies.

Expiration Tracking. The process of following the tracks or traces of or observing the moving path of expiration or termination (Merriam-Webster, n.d.-z; Merriam-Webster, n.d.-aa).

Gradient Boosting. Within this research, Gradient Boosting represents a machine learning technique that sequentially builds prediction models, with each new model correcting errors from previous models to improve forecasting accuracy for Risha Pet Supplies.

Historical Sales Data. Factual information of, relating to, or having the character of history used as a basis for reasoning, discussion, or calculation (Merriam-Webster, n.d.-ab; Merriam-Webster, n.d.-ac).

Inventory. An itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-f).

Inventory Accuracy. For the purposes of this study, inventory accuracy measures the degree of correspondence between recorded stock quantities in the system and actual physical inventory counts at Risha Pet Supplies.

Inventory Management. The conducting or supervising of an itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-f; Merriam-Webster, n.d.-g).

Machine Learning (ML). The process by which a computer is able to improve its own performance by continuously incorporating new data into an existing statistical model (Merriam-Webster, n.d.-h).

Point-of-Sale (POS). The place at which a retail transaction is carried out or a computerized system used in such places for recording sales and related functions (Merriam-Webster, n.d.-i).

Predictive Analytics. A branch of advanced analytics that makes predictions about future outcomes using historical data combined with statistical modeling, data mining techniques and machine learning (TechTarget, n.d.).

Purchase Order. Within this research, a purchase order is the order for stock that FETCH sends to a supplier, listing the products and quantities Risha Pet Supplies needs.

Random Forest. In this study, Random Forest refers to a machine learning algorithm that creates multiple decision trees using different data samples to generate

accurate demand forecasts through collective prediction averaging for Risha Pet Supplies inventory.

Real-Time Inventory. The actual time during which something takes place for an itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-x; Merriam-Webster, n.d.-f).

Reorder Point. Within this research, reorder point indicates the predetermined inventory level that triggers an automatic purchase order to replenish stock before depletion occurs at Risha Pet Supplies.

Safety Stock. For the purposes of this study, safety stock refers to the buffer inventory quantity maintained above regular demand levels to prevent stockouts during unexpected demand increases or supplier delays at Risha Pet Supplies.

Seasonal Demand. The quantity of a commodity or service wanted at a specified price and time that is of, relating to, or varying in occurrence according to the season (Merriam-Webster, n.d.-d; Merriam-Webster, n.d.-j).

Stock Keeping Unit (SKU). A number or string of alphanumeric characters that uniquely identify a product and used to track inventory in retail stores, warehouses, and online retailers (Merriam-Webster, n.d.-p).

Stockout. The depletion of stock on hand of an item offered for sale (Merriam-Webster, n.d.-l).

Supplier Notification. In this study, supplier notification refers to automated alerts sent to product vendors when Risha Pet Supplies requires inventory replenishment, streamlining the procurement process and reducing order delays.

Supplier Portal. In this study, the Supplier Portal is the part of FETCH where a supplier signs in to see and confirm its purchase orders, mark them as shipped, offer orders, request price changes, and see its payments and star rating.

Supply Chain Management. The management of the flow of goods and services from raw material sources to end customers, including procurement, production, inventory management, and distribution (Oxford Reference, n.d.-d).

System Testing. For the purposes of this study, system testing is the test in which the administrator, the cashier and the suppliers each go through every screen of their role in order, from signing in to logging out, to check that FETCH works as a whole.

Unit Testing. For the purposes of this study, unit testing is the test in which technical experts check each module of FETCH separately through completeness, format, range, consistency and database checks.

User Acceptance Testing (UAT). In this study, UAT refers to the unit testing and system testing of FETCH, carried out to check that the system meets the needs of its users and is ready to be used at Risha Pet Supplies.

Chapter 2

Methods

This chapter discusses the research methodology utilized during the development, design, and evaluation of the Intelligent Inventory Management System with Predictive Analytics for Risha Pet Supplies. This includes the explanation of the design, location, population, sample, research instrument, methods of gathering information, as well as the method of analysis to be utilized.

Research Locale

The study is to be conducted at the business establishment of Risha Pet Supplies based at 07 Bagumbong Road, Brgy. 171, Caloocan City, Metro Manila. This business caters to pet-related item sales, which focuses on products intended for dog and cat companions, with only some available products for birds. The present products include pet food, accessories, health-related products, grooming products, and many more.

Risha Pet Supplies is operating daily from 6am-5pm and serves its customers in the surrounding environs. As of now, the business engages in manual handling to control its inventory. This presents a challenge when it comes to the tracking and demand forecasting of the inventory. Risha Pet Supplies presents an excellent avenue for research in the development and testing of an intelligent inventory management system.

Research Design

In a bid to systematically develop, test, and assess FETCH, this study proposes to use particular methodologies that will ensure a systematic development process while concurrently testing its efficiency. FETCH shall adopt a developmental research design

methodology that combines the Agile SDLC software development life cycle model with a descriptive-evaluative design.

Agile Software Development Life Cycle (SDLC). In FETCH, the proposed project will adopt a developmental research design with a descriptive-evaluative method involving the Agile Software Development Life Cycle (SDLC). The former will be concerned with the development phase of the Intelligent Inventory Management System, where repeated stages of software development will be undertaken in the design of the system. The study will adopt the Agile SDLC process due to its ability to facilitate continued software development and the inclusion of feedback from stakeholders on the software system design. This is unlike the Waterfall model that involves the sequential process of software system development. The need for adaptability in software system requirements cannot be over-estimated, especially where interactions between the software developers or design teams and software end users may result in an altered software system or new requirements. The process of development is to be done using Agile methodologies such as sprint planning, coordination, sprint reviews, and retrospectives. The descriptive-evaluative component of the research aims to use quantitative approaches in order to obtain data that can reveal the effectiveness of the system's usability, as well as user satisfaction, using a questionnaire after the hands-on evaluation of the system. It provides a systematic approach to building the system, as well as a way of objectively determining the effectiveness of the system.

Figure 2

Agile Development Model



FETCH would be developed based on the following process:

Planning. The team will leverage this stage to set objectives of FETCH as well as all the requirements needed. These requirements cover server as well as database requirements, as well as requirements of software including Visual Studio Code, MySQL Workbench, as well as Python environment. The team aims at finding all the requirements needed in order to implement functionalities of inventory management, barcode scanner functionality, autoordering functionality, demand forecasting functionality, as well as expiration dates.

Design. The study will comprise the design of the system's interfaces and architecture. The design will be aimed at being user-friendly so that Risha Pet Supplies personnel can use it to control and track the stock levels and also for carrying out transactions. The designs for the dashboard of the management and administrators will entail the use of predictive analytics and notifications.

Development. The system will be developed based on the design that has been accepted. This will entail coding the front end of the system with HTML, CSS and JavaScript, coding the back end with Node.js and the Express framework connected to a MySQL database, and coding the machine learning models with Python. The system will include functionalities such as inventory management, barcode scanning, automated reordering, supplier management, and inventory forecasting with the Random Forest and Gradient Boosting algorithms.

Testing. User Acceptance Testing (UAT) will be conducted in two levels to check that the system works as it should. In unit testing, technical experts will test each module using the Unit Test Plan, which covers completeness, format, range, consistency and database checks. In system testing, the administrator, the cashier and the suppliers will each follow the System Test Plan for their role, going through every screen from signing in to logging out. Afterwards, the users will answer the Computer System Usability Questionnaire (CSUQ) to evaluate the usability of the system.

Deployment. The system is to be deployed at Risha Pet Supplies, where it will be accessible to staff and management for implementation. This stage will ensure that it is fully integrated with all hardware, software, and barcode scanners, while still incorporating real-time inventory management and replenishment.

Review. The system's efficiency and effectiveness will be measured as feedback will be obtained from stakeholders and the system's users on the system's performance and level of satisfaction with the system.

Respondents of the study

The study will select respondents through purposive sampling to ensure that participants have relevant experience in inventory operations, system usage, technical development, and supply chain processes connected to the FETCH: Inventory Management System with Predictive Analytics. This method supports the developmental and descriptive-evaluative research design by helping researchers gather reliable and practical feedback on the system's usability, technical performance, and effectiveness in real-world situations. The chosen respondents will take part in interviews, system testing, User Acceptance Testing (UAT), and usability evaluations using the Computer System Usability Questionnaire (CSUQ).

The sample size is based on established research in usability testing. Budiu (2021) explains that a qualitative usability test with five participants finds about 85% of the usability problems of an interface. For systems used by different types of users, Seinsche et al. (2023) showed that usability studies benefit from 8 to 15 participants per group, so that every type of user is represented. Caron et al. (2022) add that the sample size has to balance finding usability problems against the resources available, and Sebele-Mpofu (2020) notes that purposive sampling of information-rich participants reaches data saturation without depending on a large sample.

Following these studies, the respondents will be one (1) owner and one (1) staff member of Risha Pet Supplies, who are the main users of the system as its administrator and cashier; ten (10) suppliers, who will test the Supplier Portal, including the purchase orders and notifications sent to them; eight (8) technical experts with a background in

Computer Science, Information Technology or a related field, who will conduct the unit testing of each module and assess the system's architecture, functionality and the accuracy of its predictive analytics; and ten (10) owners or managers of other small pet shops, who will examine the system's relevance to the industry. In all, thirty (30) respondents will evaluate FETCH from operational, technical, business and supply chain perspectives. This approach makes the assessment thorough and fits the study's research methodology.

Research Instrument

To comprehensively assess FETCH's functionality and usability, the researchers will employ multiple data-gathering instruments. These tools will enable systematic evaluation of system performance, user satisfaction, and alignment with the operational requirements of Risha Pet Supplies.

Documentary Analysis. The researchers will undertake a literature analysis of research works related to the development of inventory management systems, predictive analytics, machine learning technologies in the retail industry, or automated reordering processes to use in building the FETCH system. The sources will cover local literature as well as worldwide research works for demand forecasting methods, ensemble learning approaches, real-time inventory monitoring, or point-of-sale integration. The sources of documentation shall form the basis for developing the techniques of the new system to enhance inventory accuracy, minimize inventory shortages, or maximize inventory levels of Risha Pet Supplies.

Guide Questions. The researchers will interview the owner and the manager of Risha Pet Supplies, who handle the inventory and the ordering from suppliers, to

understand the requirements of the system. The interview will focus on how the store currently tracks its stock, how it decides what to reorder and when, and what the system needs to do to be useful in their daily work. The suppliers will not be interviewed; their part of the system will be based on the ordering process described by the owner, and they will take part only in the system testing and the CSUQ.

User Acceptance Testing (UAT). The researchers will prepare test case sheets to check that FETCH works as expected. The Unit Test Plan will be used by the technical experts to test each module with completeness, format, range, consistency and database checks (see Appendix B). The System Test Plans, one each for the administrator, the cashier and the supplier, will be used by the end users to carry out their tasks screen by screen, such as adding a new product, scanning barcodes and completing sales at the point of sale, checking the demand forecast, creating purchase orders, confirming and shipping orders in the Supplier Portal, and tracking stock levels on the dashboard (see Appendices C to E). The tests will also check that the front end, the Node.js back end and the Python forecasting models work together correctly.

Computer System Usability Questionnaire (CSUQ). In this respect, the researchers will be using the "Yes"/ "No" dichotomous scale to gauge system usability, the quality of the interface, and user satisfaction. This will further enable staff and managers to give candid feedback about how easy FETCH will be to navigate, how clear the predictive analytics visualizations are, and their general feeling of confidence in using the system to execute everyday inventory management tasks.

Data Gathering Procedure

The data gathering will be facilitated by the researchers in a structured, multi-phased approach, using both quantitative records and qualitative feedback from the management of Risha Pet Supplies. First, the researchers will arrange with the owners of the store to retrieve historical sales data, inventory logs, and supplier invoices from the last three (3) to twelve (12) months. This raw data will be collected through a data extraction template and will be the main input for the training of the ensemble machine learning models, the Random Forest and Gradient Boosting algorithms. Upon analyzing these historical records, the researchers predict that they will determine sales patterns, such as seasonal fluctuations and spikes in demand on weekends, which are crucial for the predictive functions of the system.

When it comes to the planning or designing cycle of the development life cycle, semi-structured interviews, as well as observations, among the store's employees, will be conducted to ascertain the functional requirements. By using an interview guide as the research instrument, the researchers will encode the deficiencies in the store's existing manual spreadsheet process and the "intuition-based" reordering process. The understanding developed this way aims to ensure that the software interface is user-friendly, based on actual users, like the use of larger elements of the interface for the use of tablets, as determined in the designing cycle.

The researchers will finally collect evaluative data through unit testing, system testing and the CSUQ. In unit testing, the technical experts will use the Unit Test Plan to check each module, including the accuracy of the stock forecast and the reorder alerts. As

the modules are completed, the administrator, the cashier and the suppliers will follow the System Test Plan for their role and use the system as they would in their daily work. Their feedback will then be gathered through the CSUQ, which asks about the usability of the system and the reliability of the AI-driven insights. This iterative collection of data will ensure that the final system actually minimizes costs and prevents stockouts as intended by the study.

Statistical Treatment

The section below discusses the statistical techniques and tools to be utilized in the analysis of the data obtained during User Acceptance Testing (UAT) and the Computer System Usability Questionnaire (CSUQ). The techniques will test for usability, functionality, and effectiveness of the system being implemented.

Percentage-Frequency Distribution. A frequency distribution is to be employed to organize all the data that has been collected from the respondents. In percentage-frequency distribution, the total number of responses made on each criterion is represented, giving a visual idea about the distribution of feedback received.

Percentage. After the responses are tallied, each count will be turned into a percentage of all the responses, using the formula:

$$\text{Percentage (\%)} = (\text{Number of Responses in a Category} \div \text{Total Number of Responses}) \times 100$$

Dichotomous Scale. The study will use a scale of dichotomous (YES/NO response options) in measuring the usability and acceptability level of the system. This is

to ensure that the study has a flat response on the acceptability of certain features of the predictive analytics and inventory modules.

This percentage will be calculated for each item and each subscale, and it will also be calculated for system evaluation.

Table 1

Rating for Level of Usability (CSUQ)

In order to give a qualitative aspect to the obtained data, a comparison between the obtained results and a certain benchmark will be carried out. This helps to establish if the Inventory Management System is ready to be put to use or if it still needs development.

Range	Descriptive rating	Interpretation
≥ 78%	Passed (P)	The item/criteria passed the system evaluation
< 78%	Failed (F)	The item/criteria failed the system evaluation

The criteria against which evaluation of the Inventory Management System is conducted are presented below in Table 1. The passing criteria is set at 78% based on a benchmark, and all systems above this point are rated acceptable in terms of predictive and management functionality, while those below this point fall below user requirements.

Appendix A

CONSENT FORM

Dear Ma'am/Sir,

We are students of the Bachelor of Science in Computer Science at De La Salle Araneta University, currently conducting a research study entitled:

"FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies"

Our system is now live and accessible for testing. It is designed to improve inventory management efficiency and support data-driven decision-making for pet supply retail operations. The system features predictive analytics using ensemble machine learning algorithms (Random Forest and Gradient Boosting), automated reordering capabilities, barcode scanning integration, real-time inventory tracking, supplier management, and expiration monitoring. The system has undergone multiple phases of development and has been reviewed and approved by our thesis adviser.

We would like to invite you to explore our system and experience its features firsthand. Afterward, we kindly ask that you complete a brief 19-item questionnaire consisting of simple yes-or-no questions regarding the system's usability and functionality.

Your feedback will be highly valuable in improving our system. All responses will remain confidential and will be used strictly for academic purposes.

Thank you very much for your time and support.

In St. La Salle.

_____	_____
Joylinda P. Jugan	Juan Carlo M. Reusora
Researcher	**Researcher**

CONSENT APPROVAL

Evaluation Name (Optional): _____

Position / Company: _____

Signature: _____

Computer System Usability Questionnaire (CSUQ)

FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies

Please rate the usability of the system.

- Try to respond to all the items by putting a check mark (✓) on the respective column of your response.
- For items that are not applicable, use: NA
- Add a comment about an item if your response is No

Overall Reaction to the System	Yes	No	N/A	Remarks If No, what is your comment.
1. Overall, I am satisfied with how easy the system is to use.				
2. The system is easy and simple to use for managing pet supply inventory.				
3. I can complete inventory tasks (stock tracking, barcode scanning, POS transactions) effectively using the system.				
4. The system allows me to complete my work quickly with automated features.				
5. The predictive analytics and automated reordering help me perform my work efficiently.				
6. I feel comfortable when using the system for daily inventory operations.				
7. Learning to use the system's features (barcode scanning, forecasting, reordering) was easy.				
8. I became productive quickly when using the system.				
9. The system provides clear alerts (low stock, expiration warnings, reorder notifications) that help me take action.				
10. When I make mistakes, I can recover easily and continue using the system.				
11. The system provides clear and helpful information (demand forecasts, inventory reports, supplier notifications).				
12. It is easy to find inventory information, sales data, and forecasting reports I need in the system.				
13. The demand forecasting results and inventory analytics are easy to understand.				

14. The predictive analytics and automated alerts help me make better stocking decisions.				
15. The dashboard and inventory information on the screen are well organized and clear.				
16. The system's interface is pleasant to look at.				
17. I like using the system's interface for inventory management tasks.				
18. The system has all the features I expect (predictive analytics, automated reordering, barcode scanning, supplier management, expiration tracking).				
19. Overall, I am satisfied with the system.				

Appendix B

Unit Test Plan

UNIT TEST PLAN		Legend:	Passed (Result is same as expected)
Project Name:	FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies		Failed (Actual result is NOT same as expected result)
Project Manager:	Juan Carlo Reusora and Joylinda Jugan		In Progress (Test is still in progress or not yet done)
Date:			Open (Not yet tested)
Run/Tested by:			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
ALL ROLES — ADMINISTRATOR, CASHIER AND SUPPLIER								
1 LOGIN MODULE								
	1 Completeness Check							
	1	Check if a valid sign in works	1-01	Enter the username and the password, then click Sign in	The user is signed in and taken to the home page of the account			
	2	Check if Username is empty	1-02	Leave Username empty, then click Sign in	A tooltip "Please fill out this field." is displayed on Username			
	3	Check if Password is empty	1-03	Leave Password empty, then click Sign in	A tooltip "Please fill out this field." is displayed on Password			
	4	Check if the reset username or email is empty	1-04	Click "Forgot your password?", leave Username or email empty, then click Ask for a reset	A tooltip "Please fill out this field." is displayed			
	5	Check if New password is empty	1-05	Open a reset link, leave New password empty, then click Save new password	A tooltip "Please fill out this field." is displayed			
	2 Format Check							
	1	Check if a password with every kind of character is accepted	2-01	Open a reset link, enter a new password with an uppercase letter, a lowercase letter, a number and a special character, then click Save new password	A message "Password has been reset. You can now sign in. Redirecting to sign in..." is displayed			
	2	Check if the password has no uppercase letter	2-02	Enter a new password without an uppercase letter	A message "Password must contain at least one uppercase letter (A–Z)" is displayed			
	3	Check if the password has no lowercase letter	2-03	Enter a new password without a lowercase letter	A message "Password must contain at least one lowercase letter (a–z)" is displayed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	4	Check if the password has no number	2-04	Enter a new password without a number	A message "Password must contain at least one number (0–9)" is displayed			
	5	Check if the password has no special character	2-05	Enter a new password without a special character	A message "Password must contain at least one special character (e.g. !@#$%)" is displayed			
	3 Range Check							
	1	Check if a password of 6 or more characters is saved	3-01	Enter a new password of 6 characters or more that follows the rules, then click Save new password	The password is saved and the user is taken to the Sign in page			
	2	Check if the password is shorter than 6 characters	3-02	Enter a new password with fewer than 6 characters	A message "Password must be at least 6 characters long" is displayed			
	3	Check if a 6th wrong sign-in is blocked	3-03	Enter a wrong password 5 times within 15 minutes, then try a 6th time	A message "Too many login attempts. Try again in (seconds)s." is displayed and sign-in is refused			
	4 Consistency Check							
	1	Check if the Administrator opens the Dashboard	4-01	Sign in as Administrator	The Dashboard opens: the administrator's home page, with the shop's figures			
	2	Check if the Cashier opens the Point of Sale	4-02	Sign in as Cashier	The Point of Sale opens, not the Dashboard, with "Cashier: (name)" of the cashier who signed in			
	3	Check if the Supplier opens the Supplier Portal	4-03	Sign in as Supplier	The Supplier Portal opens with that supplier's own orders and products			
	4	Check if the two passwords do not match	4-04	Open a reset link, enter a new password and a different password in Confirm password, then click Save new password	A message "Passwords do not match." is displayed and the password is not changed			
	5 Database Check							
	1	Check if a saved account can sign in	5-01	Enter a username and password that are saved in the system, then click Sign in	The user is signed in			
	2	Check if "Keep me signed in" keeps the session	5-02	Tick "Keep me signed in", sign in, close the browser and open FETCH again	The user is still signed in			
	3	Check if the password is wrong	5-03	Enter a username with a wrong password, then click Sign in	A message "Invalid username or password" is displayed			
	4	Check if the email is not yet confirmed	5-04	Sign in with an account whose email is not yet confirmed	A message "Please confirm your email address first. Check (email) for the confirmation link." is displayed with a "Resend it" link			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	5	Check if the account is deactivated	5-05	Sign in with an account that was deactivated	A message "Invalid username or password" is displayed and the account cannot sign in			
	6	Check if the reset link has expired or was used	5-06	Open a reset link that has expired or was already used	A message "This reset link is invalid or has expired. Please request a new one." is displayed			
ADMINISTRATOR								
2 NAVIGATION (SIDEBAR AND TOP BAR)								
	6 Consistency Check							
	1	Check if each menu opens its own records	6-01	Sign in as Administrator and click Dashboard, Inventory, Sales, Suppliers, Reports and Users in the sidebar one after another, then click the bell and View all notifications	Each menu opens its own page with its own records: Inventory lists products, Sales lists sales, Suppliers lists suppliers and Users lists accounts; View all notifications under the bell opens Notifications with the notices			
	2	Check if the bell lists the latest notices	6-02	Click the Notifications bell	The drop-down lists the latest notices from the Notifications page, with the number of unread ones			
	3	Check if Jump to finds a product	6-03	Press Ctrl + J (Cmd + J on a Mac), type the name of a product, then press Enter	The Jump to window lists that product and opens it			
	4	Check if a cashier can open Inventory	6-04	Sign in as Cashier, then type the address of the Inventory page (inventory.html)	The page does not open: the cashier is returned to the Point of Sale			
	5	Check if a supplier can open the Dashboard	6-05	Sign in as Supplier, then type the address of the Dashboard (dashboard.html)	The page does not open: the supplier is returned to the Supplier Portal			
3 DASHBOARD								
	7 Consistency Check							
	1	Check if the product counts match Inventory	7-01	Compare Total Products, Restock Needed and Expiring Soon with the matching cards in Inventory	The numbers are the same on both pages (both count the same product records)			
	2	Check if Total Revenue matches the Sales page	7-02	Compare Total Revenue with Total Sales Amount on the Sales page	The amounts are the same (both count the same sales)			
	3	Check if "Within 30 days" lists only those products	7-03	Click the "Within 30 days" card in the Expiring products panel	Only products whose expiration date is within the next 30 days are listed			
	4	Check if Needs restocking lists only low products	7-04	Look at the Needs restocking list	Only products at or below their reorder level are listed			
	5	Check if a voided sale is left out of revenue	7-05	Void a sale, then look at Recent Sales and Total Revenue	The sale is marked "Voided" and is not counted in Total Revenue			
	6	Check if a well-stocked product is left out	7-06	Look at the Needs restocking list for a product whose stock is above its reorder level	That product is not listed			
4 INVENTORY MODULE								

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	8 Completeness Check							
	1	Check if a complete product is added	8-01	Click Add, fill in Product name and Category, choose the Product type, fill in Unit price and Supplier, then click Save product	A message "Product added" is displayed with the SKU it was given (for example DRY-121), and the product is in the list			
	2	Check if Product type is not chosen	8-02	Click Add, leave Product type unselected, then click Save product	A message "Choose the product type. The SKU is made from it." is displayed under Product type, and the cursor goes to the first field that needs fixing			
	3	Check if Product name is empty	8-03	Click Add, leave Product name empty, then click Save product	A message "Enter the product name." is displayed under Product name			
	4	Check if Category is not chosen	8-04	Click Add, leave Category unselected, then click Save product	A message "Choose a category from the list." is displayed under Category			
	5	Check if Unit price is empty	8-05	Click Add, leave Unit price empty, then click Save product	A message "Enter the unit price (the selling price)." is displayed under Unit price			
	6	Check if Supplier is not chosen	8-06	Click Add, leave Supplier unselected, then click Save product	A message "Choose a supplier. A product needs one so it can be reordered." is displayed under Supplier			
	7	Check if the stock quantity is empty	8-07	Click Stock on a product, leave the quantity empty, then click Save adjustment	A message "Enter a quantity above zero." is displayed			
	8	Check if the barcode is empty	8-08	Click Barcodes, leave a barcode empty, then press Enter	A message "Enter a barcode first" is displayed			
	9	Check if the expiration date of new stock is empty	8-09	Click Stock on a product that has expired, choose "Add stock (received, found, returned)", leave the expiration date empty, then click Save adjustment	A message "(product) expired on (date). Enter the expiration date of the stock being added." is displayed and the stock does not change			
	9 Format Check							
	1	Check if a correct price and stock are saved	9-01	Click Add, choose a Product type, enter a price such as 120.50 and a whole number in Stock on hand, then click Save product	The product is saved with those values			
	2	Check if letters can be typed in Unit price	9-02	Click Add and type letters in Unit price	The letters are not typed: Unit price takes numbers only			
	3	Check if letters can be typed in Stock on hand	9-03	Click Add and type letters in Stock on hand	The letters are not typed: Stock on hand takes numbers only			
	4	Check if the SKU can be typed in	9-04	Click Add and try to type in the SKU field	Nothing can be typed: the SKU field is read-only and is filled in from the Product type			
	5	Check if the barcode has a space	9-05	Click Add, enter a barcode with a space, for example "48 00", then leave the field	A message "The barcode can use letters, numbers and dashes only, with no spaces." is displayed under Barcode			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	6	Check if the price has more than 2 decimals	9-06	Click Add, enter 120.555 in Unit price, then leave the field	A message "Use at most 2 decimal places, for example 120.50." is displayed under Unit price			
	7	Check if stock sold by the piece has a decimal	9-07	Click Add, keep Sold by on Piece and enter 2.5 in Stock on hand, then leave the field	A message "Stock sold by the piece has to be a whole number." is displayed			
	8	Check if the photo is not an image	9-08	Click Add, click Upload and choose a file that is not an image	A message "Please choose an image file" is displayed			
	9	Check if the CSV has no sku or name column	9-09	Click Import and choose a CSV file without the sku or name column	A message "Missing required columns: sku, name" is displayed			
	10	Check if the CSV has the header row only	9-10	Click Import and choose a CSV file with the header row only	A message "CSV must have a header row and at least one data row" is displayed			
	10 Range Check							
	1	Check if removing stock within what is on hand works	10-01	Click Stock on a product with 20 in stock, choose "Remove stock (taken out)", enter 5, then click Save adjustment	A message "Recorded −5. Stock is now 15." is displayed			
	2	Check if expired stock is replaced when stock is added	10-02	Click Stock on an expired product with 20 in stock, choose "Add stock (received, found, returned)" and enter 6	The line under the quantity reads "The 20 expired will be removed. Stock will go from 20 to 6."			
	3	Check if the product name is too short	10-03	Click Add, enter A as the Product name, then leave the field	A message "The product name is too short: use at least 2 characters." is displayed			
	4	Check if Stock on hand is below 0	10-04	Click Add and enter -5 in Stock on hand, then leave the field	A message "The stock on hand cannot be less than 0." is displayed			
	5	Check if Reorder level is below 0	10-05	Click Add and enter -1 in Reorder level, then leave the field	A message "The reorder level cannot be less than 0." is displayed			
	6	Check if Cost price is above Unit price	10-06	Click Add, enter 120.50 as the Unit price and 150 as the Cost price, then leave the field	A message "The cost price (₱150.00) is higher than the unit price (₱120.50): the shop would lose money on each sale. Check both prices." is displayed			
	7	Check if Max stock level is not above Reorder level	10-07	Click Add, enter 10 as the Reorder level, open More details and enter 5 as the Max stock level	A message "The max stock level has to be higher than the reorder level (10)." is displayed			
	8	Check if Unit price is too large	10-08	Click Add and enter 100000000 in Unit price, then leave the field	A message "The unit price is too large (₱99,999,999.99 at most)." is displayed			
	9	Check if the expiration date is before today	10-09	Click Add and choose an Expiration date before today, then click Save product	A message "The expiration date cannot be earlier than today." is displayed			
	10	Check if the stock quantity is 0	10-10	Click Stock on a product, enter 0, then click Save adjustment	A message "Enter a quantity above zero." is displayed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	11	Check if more stock is removed than is on hand	10-11	Click Stock on a product with 20 in stock, choose "Remove stock (taken out)", enter 999, then click Save adjustment	A message "Only 20 in stock: you cannot remove 999." is displayed and the stock stays 20			
	12	Check if a bulk adjustment of 0 is entered	10-12	Tick two products, click Adjust stock, enter 0, then confirm	A message "Enter the quantity as a number other than 0: a positive number adds stock, a negative number removes it." is displayed			
	11 Consistency Check							
	1	Check if Low Stock lists only low products	11-01	Click the Low Stock chip	Only products at or below their reorder level are listed, and the count on the chip equals the number listed			
	2	Check if Out of Stock lists only products at 0	11-02	Click the Out of Stock chip	Only products with 0 in stock are listed			
	3	Check if Expired lists only expired products	11-03	Click the Expired chip	Only products whose expiration date has passed are listed			
	4	Check if the category and species filters match	11-04	Choose a category in All Categories and a species in All Species	Only products of that category and species are listed			
	5	Check if the search lists only matching products	11-05	Type a name, SKU or barcode in the search box	Only the products that match are listed			
	6	Check if the Restock needed card matches	11-06	Click the Restock needed card	Only products at or below their reorder level are listed			
	7	Check if the stock preview adds up	11-07	Click Stock on a product with 20 in stock and type 5	The line under the quantity reads "Stock will go from 20 to 25.", the same stock the list shows			
	8	Check if Expired leaves out unexpired products	11-08	Click the Expired chip	Products that have not expired are not listed			
	9	Check if a deleted product leaves the till	11-09	Delete a product, then look for it at the Point of Sale	The product is no longer in the till or the reorder list			
	10	Check if a search with no match says so	11-10	Type a name no product has in the search box	"No products match your filters" is displayed			
	12 Database Check							
	1	Check if a new product gets the next SKU	12-01	Click Add, choose a Product type and enter a name and a barcode no other product has, then click Save product	The product is saved with the next number of the shop's SKUs, for example DRY-121 (a number is never given twice, even after a product is deleted)			
	2	Check if a product edit is saved	12-02	Open a product, click Edit, change the Unit price, then click Save product	A message "Product updated" is displayed and the new price is shown			
	3	Check if a product is deleted	12-03	Open a product, click Delete, then click Yes, Delete	A message "Product deleted" is displayed and the product leaves the list			
	4	Check if reordering creates purchase orders	12-04	Click Reorder, tick the products, click Reorder Selected, then click Yes, Create Orders	A message "Purchase orders sent" is displayed and each supplier is emailed its order			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	5	Check if expired stock is written off	12-05	Click Stock on a product that has expired, choose "Add stock (received, found, returned)", enter the quantity and the new expiration date, then click Save adjustment	A message "Recorded +(qty). The (n) expired were removed. Stock is now (qty)." is displayed: the expired stock is written off, the new stock replaces it, and the new expiration date is set so the product can be sold again			
	6	Check if the SKU can be changed	12-06	Open a saved product, click Edit and look at the SKU	The SKU is shown but cannot be changed, and there is no Product type to choose			
	7	Check if the product name is already used	12-07	Click Add and enter a Product name another product already has, then click Save product	A question "This name is already in use" names the other product and its SKU			
	8	Check if the barcode is already used	12-08	Click Barcodes, tick Show all, type a barcode that another product already uses, then press Enter	A message "The barcode (code) is already used by (product) (SKU (sku)). Each product needs its own barcode." is displayed			
	9	Check if Reorder is clicked when nothing is low	12-09	Click Reorder when no product is at or below its reorder level	A message "No products need reordering" is displayed			
	10	Check if Save product is clicked twice	12-10	Click Save product twice quickly	The button shows "Saving…" and only one product is saved			
5 SALES HISTORY								
	13 Completeness Check							
	1	Check if a date range filters the sales	13-01	Choose a From and a To date, then click Filter	Only the sales between those dates are listed			
	2	Check if no date is chosen	13-02	Click Filter without choosing any date	A message "Select a date range" is displayed			
	3	Check if the void reason is empty	13-03	Click Void on a sale, leave the reason empty, then click Void Sale	A message "A reason is required to void a sale" is displayed			
	4	Check if the cash counted is empty	13-04	Click End of day, leave the cash counted empty, then click Save count	A message "Cash counted is not a valid amount" is displayed			
	14 Range Check							
	1	Check if a count of 0 or more is accepted	14-01	Click End of day, enter 0 or more as the cash counted	The line under it shows "Balanced" with a tick, "Over by ₱(amount)" or "Short by ₱(amount)"			
	2	Check if From is later than To	14-02	Choose a From date later than the To date, then click Filter	A message "The start date is after the end date" is displayed			
	3	Check if the cash counted is below 0	14-03	Click End of day and type -1000 as the cash counted	The line under it reads "Enter what was counted (0 or more)"			
	15 Consistency Check							
	1	Check if only the sales in the range are listed	15-01	Choose a From and a To date, then click Filter	Only the sales between those dates are listed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	2	Check if a sale number finds that sale	15-02	Type a sale number in the search box	Only that sale is listed			
	3	Check if Sale Details match the list	15-03	Click View on a sale	The Sale Details window shows the same items and total as that sale's row			
	4	Check if Total Sales Amount matches the Dashboard	15-04	Compare Total Sales Amount with Total Revenue on the Dashboard	The amounts are the same			
	5	Check if a voided sale is left out of the total	15-05	Look at a voided sale in the list	It is marked Voided and is not counted in Total Sales Amount			
	6	Check if a voided sale can be voided again	15-06	Look at a sale that is already voided	It has no Void button, so it cannot be voided twice			
	16 Database Check							
	1	Check if the sales are exported	16-01	Click Export CSV while sales are listed	The CSV file is saved and a message "Sales exported successfully" is displayed			
	2	Check if a sale is voided	16-02	Click Void on a sale, enter the reason, then click Void Sale	A message "Sale voided" is displayed, the items return to stock and the status reads Voided			
	3	Check if a sale is deleted	16-03	Open a sale, click Delete sale, then click Yes, Delete	A message "Sale deleted" is displayed; the Activity log keeps a record of it			
	4	Check if the end-of-day count is saved	16-04	Click End of day, enter the cash counted, then click Save count	The count is saved with Balanced, Over by or Short by			
	5	Check if there are no sales to export	16-05	Click Export CSV when no sale is listed	A message "No sales to export" is displayed			
6 SUPPLIERS								
	17 Completeness Check							
	1	Check if a complete supplier is added	17-01	Click Add supplier, fill in Supplier name, Contact person, Email, Phone and Payment terms, then click Save supplier	A message "Supplier added" is displayed and the supplier is in the list			
	2	Check if Supplier name is empty	17-02	Click Add supplier, leave Supplier name empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Supplier name			
	3	Check if Contact person is empty	17-03	Click Add supplier, leave Contact person empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Contact person			
	4	Check if Email is empty	17-04	Click Add supplier, leave Email empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Email			
	5	Check if Phone is empty	17-05	Click Add supplier, leave Phone empty, then click Save supplier	A tooltip "Please fill out this field." is displayed on Phone			
	6	Check if the order note is empty	17-06	In the Orders view, click Notes on an order and click Send with the note empty	A tooltip "Please fill out this field." is displayed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	7	Check if the date for an expired product is empty	17-07	In the Orders view, click Receive into stock on an order with an expired product and leave the date empty	A message "(product) (expired (date)) has already expired, so enter the new batch's expiration date" is shown; clicking Receive into stock with no date shows "Enter the new batch's expiration date: the current one has already passed." and the order is not received			
	18 Format Check							
	1	Check if a correct email and phone are saved	18-01	Click Add supplier, enter an email with @ and a phone number in digits, then click Save supplier	The supplier is saved			
	2	Check if the email has no @	18-02	Enter an email without @, then click Save supplier	A tooltip "Please include an '@' in the email address." is displayed			
	3	Check if the phone number has letters	18-03	Enter a phone number with letters, for example "abc-defg", then click Save supplier	A message "Enter the phone number in digits (at least 7)." is displayed			
	19 Range Check							
	1	Check if a phone number of 7 digits or more is saved	19-01	Enter a phone number of 7 digits or more, then click Save supplier	The supplier is saved			
	2	Check if the phone number has fewer than 7 digits	19-02	Enter a phone number of fewer than 7 digits, for example 12345, then click Save supplier	A message "Enter the phone number in digits (at least 7)." is displayed			
	3	Check if the expiration date is before today	19-03	In the Orders view, click Receive into stock and enter an expiration date earlier than today	A message "The expiration date cannot be earlier than today." is displayed			
	20 Consistency Check							
	1	Check if the Rating matches Performance	20-01	Look at the Rating column, then click Performance on the same supplier	The stars and the result (Meets the standard / Below the standard / Not rated yet) are the same in both places			
	2	Check if the order history is the supplier's own	20-02	Click Performance on a supplier	The Purchase order history lists only that supplier's orders			
	3	Check if each rating condition shows its figure	20-03	Click Performance on a supplier that has received orders	Each condition is ticked or crossed with its figure: at least 1 order received; at least 75% on time; at least 90% on time; at least 90% in full; average delivery of 7 days or less. 4 stars or more meets the standard			
	4	Check if each Orders section lists its own orders	20-04	Open the Orders view	Each section lists only its own orders: Orders on the way, Offered by suppliers, Price changes to decide, and Delivered, not yet paid			
	5	Check if the search lists only that supplier	20-05	Type a supplier's name in the search box	Only that supplier is listed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	6	Check if a late supplier is below the standard	20-06	Click Performance on a supplier whose orders all arrived after the promised date	The on-time conditions are crossed and "Below the standard: it needs 4 stars or more." is displayed			
	7	Check if a supplier with no orders is not rated	20-07	Click Performance on a supplier with no order received in the last 12 months	A message "Not rated yet: no order from this supplier was received in the last 12 months." is displayed			
	21 Database Check							
	1	Check if a new supplier is saved	21-01	Click Add supplier and enter a name that is not on the list, then click Save supplier	The supplier is saved			
	2	Check if a supplier edit is saved	21-02	Click Edit on a supplier, change the phone, then click Save supplier	A message "Supplier updated" is displayed			
	3	Check if a supplier is deleted	21-03	Click Delete on a supplier, then click Yes, Delete	A message "Supplier deleted" is displayed			
	4	Check if receiving an order adds the stock	21-04	Open the Orders view, click Receive into stock on an order, then confirm	The order is received and the stock goes up by the quantities received			
	5	Check if an order is marked paid	21-05	In "Delivered, not yet paid", click Mark paid, then click Mark paid	A message "Marked as paid." is displayed			
	6	Check if the supplier name is already used	21-06	Click Add supplier and enter a name that is already on the list, then click Save supplier	A message "A supplier with this name is already on the list." is displayed			
7 REPORTS — SALES								
	22 Completeness Check							
	1	Check if a custom period is applied	22-01	Click Custom, choose a From and a To date, then click Apply	The figures, charts and tables show only that period			
	2	Check if the custom dates are empty	22-02	Click Custom, clear both dates, then click Apply	A message "Choose both a start and an end date" is displayed			
	23 Range Check							
	1	Check if From on or before To is accepted	23-01	Click Custom, choose a From date on or before the To date, then click Apply	The report shows that period			
	2	Check if From is later than To	23-02	Click Custom, choose a From date later than the To date, then click Apply	A message "The start date is after the end date" is displayed			
	24 Consistency Check							
	1	Check if each period shows only its own sales	24-01	Click Last 7 days, Last 30 days, This month and All time	Each time, all figures, charts and tables show only the sales of that period			
	2	Check if Day, Week and Month group the same sales	24-02	Click Day, Week and Month	The revenue chart and table group the same sales by day, week or month			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	3	Check if the category lines add up to Revenue	24-03	Choose All time and look at the lines under Sales by category	"Items sold, all categories", "Sales recorded without items", "Discounts given" and "Revenue" are listed and add up to Revenue			
	4	Check if a voided sale is left out of revenue	24-04	Void a sale, then look at Revenue for that day	The voided sale is not counted			
	25 Database Check							
	1	Check if the sales report is exported	25-01	Click Export CSV for a period with sales	The CSV file is saved and a message "Sales report exported" is displayed			
	2	Check if the period has no sales to export	25-02	Click Export CSV for a period with no sales	A message "There is nothing to export for this period" is displayed			
8 REPORTS — STOCK PLANNING (PREDICTIVE ANALYTICS)								
	26 Consistency Check							
	1	Check if Stock planning shows all its parts	26-01	Click the Stock planning tab	The 30-day forecast, What to reorder, Too much stock and How good is the forecast are displayed			
	2	Check if each sales-speed filter matches	26-02	In All products, click Selling fast, Selling steadily, Selling slowly and Not selling	Each filter lists only the products with that sales speed, with the count on each			
	3	Check if the detail is that product's own	26-03	Click a product in All products	The detail shows that product's own sales history and 30-day forecast			
	4	Check if What to reorder lists only products to order	26-04	Look at the What to reorder table	Only products that need ordering are listed, each with the reason, stock on hand, on order, order-by date and suggested quantity			
	5	Check if a product with little sales history is flagged	26-05	Click a product with fewer than 14 days of sales on record	The detail says "This product has only (n) days of sales on record, so this is a cautious first estimate"			
	27 Database Check							
	1	Check if the reorder list is exported	27-01	Click Export list when products need reordering	The reorder list is saved and a message "Reorder list exported" is displayed			
	2	Check if there is nothing to reorder	27-02	Click Export list when no product needs reordering	A message "Nothing needs reordering right now" is displayed			
	3	Check if no product on the list has a supplier	27-03	Open What to reorder when no product on the list has a supplier	The Create purchase orders button is not shown, so no order can be made			
9 REPORTS — VOID REQUESTS AND ACTIVITY LOG								
	28 Consistency Check							
	1	Check if only the void requests are listed	28-01	Click the Void requests tab	Waiting for a decision lists only the sales cashiers asked to void, with the reason and amount			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	2	Check if a requested sale is voided	28-02	Click Void sale on a request, then confirm	"Sale voided and its items returned to stock" is displayed and the request leaves Waiting for a decision			
	3	Check if a requested sale is kept	28-03	Click Keep sale on a request, then confirm	"Request dismissed. The sale stands." is displayed and the request leaves Waiting for a decision			
	4	Check if the activity log filters match	28-04	Click the Activity log tab, then choose an action and an area	Only the records of that action and that area are listed			
	5	Check if there are no void requests	28-05	Open Waiting for a decision when no cashier has asked for a void	No request is listed			
10 NOTIFICATIONS								
	29 Consistency Check							
	1	Check if Low Stock lists only low-stock notices	29-01	Choose Low Stock in All Types	Only low-stock notices are listed			
	2	Check if Unread lists only unread notices	29-02	Choose Unread in All Status	Only unread notices are listed			
	3	Check if a notice is marked read	29-03	Click Mark as read (the check button) beside an unread notice	It is marked read and the unread count goes down by one			
	4	Check if Mark all read marks every notice	29-04	Click Mark all read	A message "All notifications marked as read" is displayed and the unread count is 0			
	5	Check if Unread is empty after Mark all read	29-05	Choose Unread in All Status after clicking Mark all read	No notice is listed			
11 USERS (ADMINISTRATOR)								
	30 Completeness Check							
	1	Check if a complete user is created	30-01	Click Add user, enter the email, click Send code, enter the 6-digit code and every other field, then click Create user	A message "User created" is displayed and the user is in the list			
	2	Check if Full name is empty	30-02	Click Add user, leave Full name empty, then click Create user	A tooltip "Please fill out this field." is displayed on Full name			
	3	Check if Username is empty	30-03	Leave Username empty, then click Create user	A tooltip "Please fill out this field." is displayed on Username			
	4	Check if Email is empty	30-04	Leave Email empty, then click Send code	A message "Email required" is displayed with "Type the new user's email address first, then send the code."			
	5	Check if Password is empty	30-05	Leave Password empty, then click Create user	A tooltip "Please fill out this field." is displayed on Password			
	6	Check if the supplier is not chosen	30-06	Choose the role Supplier without choosing a supplier, then click Create user	A message "Supplier required" is displayed			
	31 Format Check							

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	1	Check if a correct email and password are accepted	31-01	Enter an email with @ and a password with an uppercase letter, a number and a special character	The account can be created			
	2	Check if the email has no @	31-02	Enter an email without @	A tooltip "Please include an '@' in the email address." is displayed			
	3	Check if the password has no uppercase letter	31-03	Enter a password without an uppercase letter, then click Create user	A message "Password too weak" is displayed with "Add at least one uppercase letter (A–Z)."			
	4	Check if the password has no number	31-04	Enter a password without a number, then click Create user	A message "Password too weak" is displayed with "Add at least one number (0–9)."			
	5	Check if the password has no special character	31-05	Enter a password without a special character, then click Create user	A message "Password too weak" is displayed with "Add at least one special character (e.g. !@#$%)."			
	32 Range Check							
	1	Check if a password of 6 or more characters is accepted	32-01	Enter a password of 6 characters or more	The password is accepted			
	2	Check if the password is shorter than 6 characters	32-02	Enter a password with fewer than 6 characters, then click Create user	A tooltip "Please lengthen this text to 6 characters or more (you are currently using (number) characters)." is displayed on Password			
	3	Check if the code is wrong too many times	32-03	Enter a wrong code 5 times, then try once more	The 5th wrong code shows "That code is not correct. Send a new code." and the next try shows "Too many incorrect attempts. Send a new code."			
	33 Consistency Check							
	1	Check if the search lists only matching users	33-01	Type a name in the search box	Only the matching users are listed			
	2	Check if Show deactivated adds those accounts	33-02	Tick Show deactivated	Deactivated accounts are added to the list, each with a Restore button			
	3	Check if deactivated accounts are hidden	33-03	Untick Show deactivated	Deactivated accounts are not listed			
	4	Check if you can deactivate your own account	33-04	Look at your own row in the Users list	There is no Deactivate button on your own account			
	5	Check if the email is changed after the code is sent	33-05	Send a code, then change the email address before creating the user	The line "Step 1: press "Send code" to email a code to that address." is displayed and Create user cannot be clicked			
	34 Database Check							

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	1	Check if a new user is saved	34-01	Enter a username and an email no other account uses, with the code from the email	The user is created			
	2	Check if an account is restored	34-02	Tick Show deactivated, then click Restore on a deactivated user	A message "Account restored" is displayed			
	3	Check if an account is deactivated	34-03	Click Deactivate on another user, then click Yes, deactivate	A message "Account deactivated" is displayed and that user can no longer sign in			
	4	Check if the username is already taken	34-04	Enter a username that is already taken	A message "Username already taken" is displayed			
	5	Check if the email is already used	34-05	Enter an email that another account already uses	A message "That email address is already used by (username)." is displayed			
	6	Check if the code is wrong	34-06	Enter a wrong 6-digit code, then click Create user	A message "That code is not correct. (n) attempt(s) left." is displayed			
	7	Check if the code has expired	34-07	Enter a code that has expired	A message "That code has expired. Send a new one." is displayed			
12 SETTINGS								
	35 Completeness Check							
	1	Check if a complete profile is saved	35-01	In Profile, fill in Full name and Email, then click Save changes	A message "Profile updated" is displayed			
	2	Check if Full name is empty	35-02	In Profile, clear Full name, then click Save changes	A message "Please enter your full name" is displayed			
	3	Check if Current password is empty	35-03	In Password, leave Current password empty, then click Update password	A message "Enter current password" is displayed			
	4	Check if New password is empty	35-04	In Password, leave New password empty, then click Update password	A message "Enter new password" is displayed			
	5	Check if Confirm new password is empty	35-05	In Password, leave Confirm new password empty, then click Update password	A message "Confirm new password" is displayed			
	36 Format Check							
	1	Check if a correct email is saved	36-01	In Profile, enter an email with @, then click Save changes	The profile is saved			
	2	Check if the email has no @	36-02	In Profile, enter an email without @, then click Save changes	A message "Please enter a valid email address" is displayed			
	3	Check if the new password is too weak	36-03	In Password, enter a new password without a number or special character	A message "Password too weak" is displayed with the missing rule			
	4	Check if the photo is not an image	36-04	In Profile, upload a file that is not an image as the photo	A message "Please choose an image file" is displayed (a damaged picture shows "That image could not be loaded")			
	37 Range Check							

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	1	Check if a discount limit from 0 to 100 is saved	37-01	In Preferences, enter 10 in "Largest discount a cashier can give (%)", then click Save limit	A message "Cashiers can now discount up to 10%" is displayed			
	2	Check if the discount limit is above 100	37-02	Enter 150 in "Largest discount a cashier can give (%)", then click Save limit	A message "Enter a limit between 0 and 100" is displayed			
	38 Consistency Check							
	1	Check if Session shows the signed-in user	38-01	Click the Session tab	The full name, role and account status of the signed-in user are displayed (the Cashier role reads "Staff")			
	2	Check if Profile shows the signed-in user	38-02	Click the Profile tab	The full name and email of the signed-in user are displayed			
	3	Check if the two passwords do not match	38-03	In Password, enter a different password in Confirm new password	A message "Passwords do not match" is displayed			
	39 Database Check							
	1	Check if a new email is saved	39-01	In Profile, enter an email no other account uses, then click Save changes	The profile is saved			
	2	Check if the appearance is saved	39-02	In Appearance, choose a mode, accent colour, font and text size, then click Save changes	A message "Appearance saved" is displayed			
	3	Check if the password is changed	39-03	In Password, enter the current password and the same new password twice, then click Update password	A message "Password changed" is displayed			
	4	Check if the current password is wrong	39-04	In Password, enter a wrong current password, then click Update password	A message "Password not changed" is displayed with "Current password is incorrect"			
	5	Check if the email is already used	39-05	In Profile, enter an email another account uses	A message "That email is already used by another account" is displayed			
CASHIER								
13 POINT OF SALE (CASHIER)								
	40 Completeness Check							
	1	Check if the starting cash is saved	40-01	On the first sign-in of the day, enter 500 as the Starting cash, then click Save starting cash	A message "Starting cash saved: ₱500.00" is displayed			
	2	Check if Starting cash is empty	40-02	Leave Starting cash empty, then click Save starting cash	A message "Enter the cash in the drawer, or 0 if it is empty" is displayed			
	3	Check if the cart is empty	40-03	Click Charge with an empty cart	A message "Cart is empty" is displayed			
	4	Check if Cash received is empty	40-04	Add an item, leave Cash received empty, then click Charge	A message "Enter the cash amount received" is displayed			
	5	Check if the cash amount is empty	40-05	Click Cash in / out, leave Amount empty, then click Record cash taken out	A message "Enter the amount" is displayed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	6	Check if the reason for cash taken out is empty	40-06	Choose Taken out, enter an amount and leave "Who took it, and what for" empty	A message "Say who took the cash and what for" is displayed			
	7	Check if the cash counted is empty	40-07	Click End of day, leave the cash counted empty, then click Save count and close drawer	A message "Enter the cash counted in the drawer" is displayed			
	8	Check if the void reason is empty	40-08	In My sales today, click Ask for void and leave the reason empty	A message "Say briefly what went wrong" is displayed			
	41 Format Check							
	1	Check if numbers in cash and discount are worked out	41-01	Type numbers in Cash received and Discount	The change and the total are worked out			
	2	Check if letters can be typed in Cash received	41-02	Type letters in Cash received	The letters are not typed: the field takes numbers only			
	3	Check if letters can be typed in Discount	41-03	Type letters in Discount	The letters are not typed: the field takes numbers only			
	42 Range Check							
	1	Check if enough cash is received	42-01	Enter Cash received equal to or more than the total, then click Charge	The Confirm cash sale window shows the change			
	2	Check if the quantity is more than the stock	42-02	Change an item's quantity in the cart to more than its "(n) left"	A message "Only (n) in stock" is displayed and the quantity is set to the (n) in stock			
	3	Check if the cash received is less than the total	42-03	Enter less Cash received than the total, then click Charge	A message "Insufficient cash — customer gave less than the total" is displayed			
	4	Check if the discount is above the limit	42-04	Enter a discount above the limit, for example 50	A message "The largest discount allowed is (limit)%" is displayed and the discount is lowered to the limit			
	5	Check if an expired product is scanned	42-05	Scan a product whose expiration date has passed	A message "(product) has expired and cannot be sold" is displayed			
	6	Check if a sold-out product is scanned	42-06	Scan a product with no stock	A message "(product) is sold out" is displayed			
	7	Check if more cash is taken out than the drawer holds	42-07	Take out more cash than the drawer holds	A message "That is more cash than the drawer should hold." is displayed			
	43 Consistency Check							
	1	Check if the product card matches Inventory	43-01	Look at a product card	Its price and "(n) left" are the same as the product's Unit price and stock in Inventory			
	2	Check if a category shows only its products	43-02	Click a category button	Only the products of that category are shown			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	3	Check if the search shows only matching products	43-03	Type a product name in the search box	Only the matching products are shown			
	4	Check if the card updates by itself	43-04	Change a product's stock in Inventory while the till is open, then wait 10 seconds	The product's card shows the new "(n) left" without reloading the page			
	5	Check if the subtotal and total add up	43-05	Add two items and a discount	The subtotal is the sum of the lines, and the total is the subtotal less the discount			
	6	Check if a sale lowers the stock	43-06	Complete a sale of 2 of a product, then look at its card and at Inventory	The card's "(n) left" has gone down by 2, and Inventory shows the same new stock			
	7	Check if the receipt matches the sale	43-07	Look at the receipt after Complete sale	The items, quantities, discount, total, cash received and change are the same as in the Confirm cash sale window			
	8	Check if the sale matches in Sales	43-08	Sign in as Administrator and open the same sale in Sales	It shows the same items, total and cashier as the receipt			
	9	Check if My sales today shows only this cashier's sales	43-09	Click My sales today	Only this cashier's sales of today are listed			
	10	Check if the cash in and out totals add up	43-10	Click Cash in / out	The totals line reads "Put in ₱(amount) · Taken out ₱(amount)" for this drawer today			
	11	Check if the expected cash adds up	43-11	Click End of day, type the counted cash, then save the count	The expected cash is not shown before saving (a blind count); after saving it equals starting cash + cash sales + cash put in − cash taken out			
	12	Check if "Can't be sold" lists only those products	43-12	Click "Can't be sold"	Only expired and sold-out products are listed, each marked "Expired (date)" or "None left"			
	13	Check if an unsellable product is among the cards	43-13	Look for an expired or sold-out product among the product cards	It is not among the cards; it is listed under "Can't be sold"			
	14	Check if a deleted product leaves the till	43-14	Delete a product in Inventory while the till is open, then wait 10 seconds	The product's card disappears from the till			
	15	Check if another cashier's sales are listed	43-15	Click My sales today after another cashier has made sales	Only your own sales are listed, not the other cashier's			
	16	Check if a voided sale is left out of the expected cash	43-16	Void one of today's sales (administrator), then click End of day at the till	The voided sale is not counted in the expected cash			
	44 Database Check							
	1	Check if a barcode on record adds the product	44-01	Scan the barcode of a product on record	The product is added to the cart			
	2	Check if a sale is completed	44-02	Add items, enter Cash received, click Charge, then click Complete sale	A message "Sale completed! Change: ₱(amount)" is displayed, the stock goes down and the receipt is shown			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	3	Check if cash taken out is recorded	44-03	Click Cash in / out, choose Taken out, enter the amount and reason, click Record cash taken out, then Record it	A message "Recorded" is displayed and the expected cash goes down by that amount			
	4	Check if the end-of-day count is saved	44-04	Click End of day, enter the cash counted, then click Save count and close drawer	"Balanced" with a tick, "Over by ₱(amount)" or "Short by ₱(amount)" is shown and the count is saved			
	5	Check if the barcode is not on record	44-05	Scan or type a barcode that is not on record	A message "No product found for "(barcode)"" is displayed			
	6	Check if a void can be asked for twice	44-06	Look at a sale that already has a void request in My sales today	The Ask for void button is not shown			
	7	Check if cash can be recorded after the count	44-07	After the count is saved, click Cash in / out	The window says "Today's cash count is already recorded, so nothing more can be added for today."			
	8	Check if Complete sale is clicked twice	44-08	Click Complete sale twice quickly	Only one sale is recorded and the stock goes down once			
	9	Check if a sale can be made after the count	44-09	After the count is saved, click a product card	A message "Your drawer is closed for today" is displayed			
SUPPLIER								
14 SUPPLIER PORTAL								
	45 Completeness Check							
	1	Check if an order is confirmed	45-01	Click Confirm order on a waiting order, choose the delivery date, then click Confirm order	A message "Order confirmed." is displayed			
	2	Check if the delivery date is empty	45-02	Click Confirm order and leave the delivery date empty	A message "Choose the date you will deliver" is displayed			
	3	Check if the new delivery date is empty	45-03	Click Change delivery date and leave the new date empty, then click Save new date	A message "Choose the new delivery date" is displayed			
	4	Check if the reason for the new date is empty	45-04	Enter the new date but leave the reason empty, then click Save new date	A message "Say briefly why the date changed" is displayed			
	5	Check if the offer has no quantity	45-05	Click Offer an order without entering any quantity, then click Send offer	A message "Enter a quantity for at least one product" is displayed			
	6	Check if the offer's delivery date is empty	45-06	Enter a quantity but leave the delivery date empty, then click Send offer	A message "Choose the date you could deliver" is displayed			
	7	Check if the note is empty	45-07	Click Notes and click Send with the note empty	A tooltip "Please fill out this field." is displayed			
	46 Format Check							
	1	Check if a correct receipt number is accepted	46-01	Click Mark as shipped, enter a receipt number with letters and numbers, then click Mark as shipped	A message "Order marked as shipped." is displayed			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	2	Check if the receipt number has * or @	46-02	Enter a receipt number with * or @	A message "The receipt number can use letters, numbers, spaces and . / # _ - only." is displayed			
	3	Check if the phone number has letters	46-03	Click Edit contact details and enter a phone number with letters, then click Save details	A message "Enter a phone number using digits, spaces and + ( ) - only." is displayed			
	47 Range Check							
	1	Check if a price above zero is sent	47-01	Click Change price, enter a New price greater than zero, then click Send to the shop	A message "Sent to the shop for approval. The current price stays until they agree." is displayed			
	2	Check if the price is 0	47-02	Click Change price and enter 0	A message "Enter a price greater than zero" is displayed			
	3	Check if the delivery date is in the past	47-03	Click Confirm order and type a delivery date in the past	A message "The delivery date cannot be in the past." is displayed			
	4	Check if the delivery date is over a year away	47-04	Click Confirm order and choose a date more than a year away	A message "That delivery date is more than a year away." is displayed			
	5	Check if 5 offers are already waiting	47-05	Click Offer an order when 5 offers are already waiting	A message "You already have 5 offers waiting for the shop. Wait for an answer before sending more." is displayed			
	48 Consistency Check							
	1	Check if the portal lists only this supplier's orders	48-01	Sign in as Supplier	The portal lists only this supplier's orders			
	2	Check if only this supplier's products are listed	48-02	Look at Products you supply	Only the products the shop buys from this supplier are listed			
	3	Check if the stars match the shop's	48-03	Look at the star rating on the portal	It shows the same stars and conditions as the shop's Performance window, and whether it meets the standard (4 stars or more)			
	4	Check if another supplier's order is listed	48-04	Look for an order the shop placed with another supplier	It is not listed			
	5	Check if a confirmed order can be confirmed again	48-05	Look at an order that is already confirmed	It has no Confirm order button			
	6	Check if a supplier with no orders is not rated	48-06	Open the portal when no order was received in the last 12 months	A message "Not rated yet: no order from this supplier was received in the last 12 months." is displayed			
	49 Database Check							
	1	Check if a price proposal reaches the shop	49-01	Click Change price and enter a price different from the current one	The proposal is sent to the shop			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	2	Check if the contact details are saved	49-02	Click Edit contact details, change them, then click Save details	A message "Your details were updated." is displayed			
	3	Check if the price is the current one	49-03	Click Change price and enter the current price	A message "That is already the current price." is displayed			

How to read this plan

In each check, the first test case is done the right way and works; the test cases after it are done the wrong way and the system stops them. TEST CASE says what the test is for. DESCRIPTION is what the user does. EXPECTED RESULT is what the system shows or does.

The five checks

- Completeness check: a required field is left empty.
- Format check: the value is the wrong type or pattern, such as letters in a number field, an email without @, a phone number with letters, a file that is not a picture, or a password missing a kind of character.
- Range check: the value is too small, too large, too short or outside the allowed dates or limits, such as a password under 6 characters, a price of 0, a quantity above the stock, a date before today or a From date after the To date.
- Consistency check: the data shown is consistent. Each menu, tab, filter, chip or card shows exactly the records that belong there and none that do not (Low Stock lists only low-stock products, a supplier sees only its own orders, a voided sale is left out of the totals), and the same figure is the same on every page (Total Revenue on the Dashboard equals Total Sales Amount on Sales). Two fields that must agree, such as a password and its confirmation, are also checked here.
- Database check: what is saved is stored and read back correctly (a product, sale, supplier or user that is added, changed or removed), and a value is compared with what is already saved, such as a SKU, barcode, username or supplier name already in use, a wrong password or code, or a link that has expired.

Words in brackets, such as (product) or (amount), stand for the value on the screen. A tooltip is the small box the web browser shows beside a field that must be filled in; its wording comes from the browser.

Appendix C

System Test Plan (Administrator)

SYSTEM TEST PLAN — ADMINISTRATOR		Legend:	Passed (Result is same as expected)
Project Name:	FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies		Failed (Actual result is NOT same as expected result)
Project Manager:	Juan Carlo Reusora and Joylinda Jugan		In Progress (Test is still in progress or not yet done)
Date:			Open (Not yet tested)
Run/Tested by:			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
ADMINISTRATOR								
A SIGN IN								
	A.1		ST-001	Open FETCH in the browser	The Sign in page is displayed with Username, Password, "Keep me signed in" and "Forgot your password?"			
	A.2		ST-002	Enter the username and the password, then click the "Sign in" button	Redirects to the Dashboard, the Administrator's home page			
	A.2.1		ST-003	Click the eye icon in the Password field, then click it again	The password is shown as plain text, then hidden again			
	A.2.2		ST-004	Tick "Keep me signed in", then click the "Sign in" button	Redirects to the home page; after the browser is closed and opened again, the user is still signed in			
	A.3		ST-005	Click "Forgot your password?"	The Reset your password window is displayed, with the username already filled in			
	A.3.1		ST-006	Enter the username or email, then click the "Ask for a reset" button	The message "If that account exists: an administrator account is emailed a reset link straight away. Any other account waits for an administrator to approve it, and is then emailed the link." is displayed			
	A.3.2		ST-007	Click the "Cancel" button	Closes the window and returns to the Sign in page			
	A.3.3		ST-008	Open the reset link from the email	The Choose a new password page is displayed			
	A.3.4		ST-009	Enter and confirm the new password, then click the "Save new password" button	"Password has been reset. You can now sign in. Redirecting to sign in..." is displayed, then redirects to the Sign in page			
	A.4		ST-010	Open the confirmation link sent to a new account's email	The page says the address is confirmed and the account can now sign in			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
B DASHBOARD								
	B.1		ST-011	Sign in as Administrator	The Dashboard is displayed with Total Products, Total Revenue, Restock Needed, Expiring Soon and Inventory value, the sales chart, recent sales, products to restock, expiring products and purchase orders			
	B.2		ST-012	Click the "Within 30 days" card in Expiring products	A window lists the products expiring within 30 days, with their expiration date and quantity			
	B.2.1		ST-013	Click the "Close" button	Closes the window and returns to the Dashboard			
	B.3		ST-014	Click the status button of a purchase order in Purchase orders, then confirm	The purchase order moves to its next status			
	B.4		ST-015	Click "Jump to…" (or press Ctrl + J), type the name of a product, then press Enter	Opens the chosen product			
	B.5		ST-016	Click the FETCH Assistant & Help button	The assistant window opens			
	B.5.1		ST-017	Type a question, then send it	The assistant's answer is displayed			
	B.6		ST-018	Click the Notifications bell	A drop-down lists the latest notifications			
	B.6.1		ST-019	Click "View all notifications →"	Redirects to the Notifications page			
	B.7		ST-020	Click the theme icon	The screen switches between light and dark mode			
	B.8		ST-021	Click the collapse button of the sidebar, then click it again	The sidebar shows icons only, then the full sidebar again			
C INVENTORY								
	C.1		ST-022	Click "Inventory" in the sidebar	The Inventory page is displayed with the summary cards and the product list			
	C.2		ST-023	Click "Card view", then "Table view"	The products are shown as cards, then as a table			
	C.3		ST-024	Click the status chips "Low stock", "Out of stock", "Expiring soon", "Expired", "Healthy", then "All"	Each chip lists only the products with that status; "All" lists every product again			
	C.4		ST-025	Click the "Add" button	The Add new product window is displayed			
	C.4.1		ST-026	Enter the Product name, choose the Category and the Product type, enter the Unit price, Stock on hand and Reorder level, and choose the Supplier	The SKU fills in by itself from the product type (for example DRY-121); each field is checked when it is left			
	C.4.2		ST-027	Click the "Generate" button beside Barcode	An in-store barcode is filled in			
	C.4.3		ST-028	Click "More details (optional)", click the "Upload" button and choose a picture	The extra fields are displayed, and the picture's preview is shown			
	C.4.4		ST-029	Click the "Save product" button	"Product added" is displayed with the SKU given; the window closes and the product is in the list			
	C.4.5		ST-030	Click "Add", then the "Cancel" button	Closes the window without saving and returns to Inventory			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	C.5		ST-031	Click a product in the list	The Product details window is displayed			
	C.5.1		ST-032	Click the "Edit" button, change the Unit price, then click the "Save product" button	"Product updated" is displayed and the new price is shown			
	C.5.2		ST-033	Click the "Delete" button, then the "Yes, Delete" button	"Product deleted" is displayed; the product leaves the list			
	C.5.3		ST-034	Click the "Delete" button, then the "Cancel" button	The question closes and the product stays in the list			
	C.5.4		ST-035	Click the "Close" button	Closes the window and returns to Inventory			
	C.6		ST-036	Click "Stock" on a product	The Adjust stock window is displayed			
	C.6.1		ST-037	Choose "Add stock (received, found, returned)", enter the quantity, then click the "Save adjustment" button	"Stock adjusted" is displayed with "Recorded +(qty). Stock is now (n)."			
	C.6.2		ST-038	Choose "Remove stock (taken out)", enter the quantity, then click the "Save adjustment" button	"Stock adjusted" is displayed with "Recorded −(qty). Stock is now (n)."			
	C.6.3		ST-039	Click the "Cancel" button	Closes the window; the stock does not change			
	C.7		ST-040	Click the "Reorder" button	A window lists the products at or below their reorder level			
	C.7.1		ST-041	Tick the products, then click "Reorder Selected"	The "Final check" question is displayed			
	C.7.2		ST-042	Click the "Yes, Create Orders" button	"Purchase orders sent" is displayed; each supplier is emailed its order			
	C.7.3		ST-043	Click "Reorder Selected" again, then the "Cancel" button	The question closes and no purchase order is created			
	C.8		ST-044	Tick the checkboxes of two products	The bar with "Bulk reorder", "Adjust stock", "Delete" and "Clear" is displayed			
	C.8.1		ST-045	Click "Adjust stock", enter a quantity, then confirm	"Stock adjusted" is displayed with how many of the selected products were updated			
	C.8.2		ST-046	Click the "Clear" button	The ticks are removed and the bar closes			
	C.9		ST-047	Click the "Import" button	The Import Products from CSV window is displayed			
	C.9.1		ST-048	Click the "Download template" button	The CSV template is saved to the computer			
	C.9.2		ST-049	Choose a filled CSV file, then click the "Import products" button	"Import complete" is displayed with how many products were added			
	C.9.3		ST-050	Click the "Cancel" button	Closes the window and returns to Inventory			
	C.10		ST-051	Click the "Barcodes" button, scan or type a product's barcode, then press Enter	The barcode is saved for that product			
	C.10.1		ST-052	Click the "Done" button	Closes the window and returns to Inventory			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	C.11		ST-053	Click the "Images" button and upload a picture for a product	The picture is saved for that product			
	C.11.1		ST-054	Click the "Done" button	Closes the window and returns to Inventory			
D SALES								
	D.1		ST-055	Click "Sales" in the sidebar	The Sales page is displayed with the summary cards and the list of sales			
	D.2		ST-056	Choose a From and a To date, then click the "Filter" button	Only the sales between those dates are listed			
	D.2.1		ST-057	Click the "Clear" button	Every sale is listed again			
	D.3		ST-058	Click "View" on a sale	The Sale Details window is displayed with the items, totals, cashier and payment			
	D.3.1		ST-059	Click the "Print receipt" button	The receipt is displayed for printing			
	D.3.2		ST-060	Click the "Delete sale" button, then the "Yes, Delete" button	"Sale deleted" is displayed and the sale leaves the list			
	D.3.3		ST-061	Click the "Delete sale" button, then the "Cancel" button	The question closes and the sale stays			
	D.3.4		ST-062	Click the "Close" button	Closes the window and returns to Sales			
	D.4		ST-063	Click "Void" on a sale, enter the reason, then click the "Void Sale" button	"Sale voided" is displayed; the items return to stock and the sale is marked Voided			
	D.4.1		ST-064	Click "Void" on a sale, then the "Cancel" button	Closes the window; the sale is not voided			
	D.5		ST-065	Click the "Export CSV" button	The CSV file is saved and "Sales exported successfully" is displayed			
	D.6		ST-066	Click the "End of day" button	The End of day cash count (whole shop) window is displayed with the expected cash			
	D.6.1		ST-067	Enter the cash counted, then click the "Save count" button	The count is saved with Balanced, Over by or Short by			
	D.6.2		ST-068	Click the "Cancel" button	Closes the window without saving and returns to Sales			
E SUPPLIERS								
	E.1		ST-069	Click "Suppliers" in the sidebar	The Suppliers page is displayed with each supplier's stars in the Rating column			
	E.2		ST-070	Click the "Add supplier" button	The Add New Supplier window is displayed			
	E.2.1		ST-071	Fill in Supplier name, Contact person, Email, Phone and Payment terms, then click the "Save supplier" button	"Supplier added" is displayed and the supplier is in the list			
	E.2.2		ST-072	Click "Add supplier", then the "Cancel" button	Closes the window without saving and returns to Suppliers			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	E.3		ST-073	Click "Edit" on a supplier, change the phone, then click the "Save supplier" button	"Supplier updated" is displayed			
	E.4		ST-074	Click "Delete" on a supplier, then the "Yes, Delete" button	"Supplier deleted" is displayed and the supplier leaves the list			
	E.4.1		ST-075	Click "Delete" on a supplier, then the "Cancel" button	The question closes and the supplier stays			
	E.5		ST-076	Click "Performance" (or the stars) of a supplier	The star rating with each condition ticked or crossed, and the purchase order history, are displayed			
	E.5.1		ST-077	Click the close (×) button	Closes the window and returns to Suppliers			
	E.6		ST-078	Click "Emails" of a supplier	The Email history window is displayed with the read receipts			
	E.6.1		ST-079	Click the "Close" button	Closes the window and returns to Suppliers			
	E.7		ST-080	Click the "Orders" view	Orders on the way, Offered by suppliers, Price changes to decide, and Delivered, not yet paid are displayed			
	E.7.1		ST-081	Click "Mark confirmed" on a waiting order, then "Yes, update"	The order is marked confirmed			
	E.7.2		ST-082	Click "Mark shipped" on the order, then "Yes, update"	The order is marked on its way			
	E.7.3		ST-083	Click "Receive into stock", enter the new batch's expiration date, then click "Receive into stock"	The order is received and the quantities are added to stock			
	E.7.4		ST-084	Click "Cancel order" on an order, then the "Cancel order" button	The order is marked cancelled			
	E.7.5		ST-085	Click "Accept offer" on a supplier's offer, then "Accept offer"	The offer becomes a confirmed order			
	E.7.6		ST-086	Click accept on a proposed price, then "Accept price"	"Price updated in inventory." is displayed			
	E.7.7		ST-087	Click "Mark paid" on a delivered order, then "Mark paid"	"Marked as paid." is displayed			
	E.7.8		ST-088	Click "Notes" on an order, type a note, then click "Send"	The note is added under the order			
	E.7.9		ST-089	Click the "Refresh" button	The orders are loaded again			
F REPORTS								
	F.1		ST-090	Click "Reports" in the sidebar	The Sales report is displayed			
	F.2		ST-091	Click "Last 7 days", "Last 30 days", "This month" and "All time"	The figures, charts and tables change to each period			
	F.3		ST-092	Click "Custom", choose a From and a To date, then click the "Apply" button	The report shows only that period			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	F.4		ST-093	Click "Day", "Week" and "Month"	The revenue chart and table are grouped by day, week or month			
	F.5		ST-094	Click the "Export CSV" button	The report is saved and "Sales report exported" is displayed			
	F.6		ST-095	Click the "Stock planning" tab	The 30-day forecast, What to reorder, Too much stock and How good is the forecast are displayed			
	F.6.1		ST-096	Click the "Refresh" button	The forecast and the stock planning figures are worked out again			
	F.6.2		ST-097	Click the "Export list" button	The reorder list is saved and "Reorder list exported" is displayed			
	F.6.3		ST-098	Click "Selling fast", then click a product in All products	Only fast-selling products are listed; the product's sales history and forecast are displayed			
	F.6.4		ST-099	Click the "Close" button	Closes the product detail and returns to Stock planning			
	F.7		ST-100	Click the "Void requests" tab	The sales cashiers asked to void are listed with the reason			
	F.7.1		ST-101	Click "Void sale" on a request, then confirm	"Sale voided and its items returned to stock" is displayed			
	F.7.2		ST-102	Click "Keep sale" on a request, then confirm	"Request dismissed. The sale stands." is displayed			
	F.8		ST-103	Click the "Activity log" tab, then choose an action and an area	Only the records of that action and area are listed			
	F.8.1		ST-104	Click the "Refresh" button	The activity log is loaded again			
G NOTIFICATIONS								
	G.1		ST-105	Click the Notifications bell, then "View all notifications →"	The Notifications page is displayed			
	G.2		ST-106	Choose a type in All Types and a status in All Status	Only those notifications are listed			
	G.3		ST-107	Click a notification	It is marked read and the unread count goes down			
	G.4		ST-108	Click the "Mark all read" button	"All notifications marked as read" is displayed			
	G.5		ST-109	Click the "Clear read" button, then confirm	"Notifications cleared" is displayed			
	G.6		ST-110	Click the "Refresh" button	The notifications are loaded again			
H USERS								
	H.1		ST-111	Click "Users" in the sidebar	The list of users is displayed			
	H.2		ST-112	Click the "Add user" button	The Add New User window is displayed			
	H.2.1		ST-113	Enter the email, then click the "Send code" button	A 6-digit code is emailed to that address			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	H.2.2		ST-114	Enter the code, Full name, Username, Role and Password, then click the "Create user" button	"User created" is displayed and the user is in the list			
	H.2.3		ST-115	Click "Add user", then the "Cancel" button	Closes the window without saving and returns to Users			
	H.3		ST-116	Click "Edit" on a user, change the details, then click "Update user"	The user's details are saved			
	H.4		ST-117	Click "Deactivate" on another user, then the "Yes, deactivate" button	"Account deactivated" is displayed; that user can no longer sign in			
	H.5		ST-118	Tick "Show deactivated", then click "Restore" on the user and confirm	"Account restored" is displayed			
	H.6		ST-119	Click "Approve and email link" on a password reset request	The request is approved and the reset link is emailed			
I SETTINGS								
	I.1		ST-120	Click the Settings (gear) icon	The Profile tab is displayed			
	I.1.1		ST-121	Change the full name, then click the "Save changes" button	"Profile updated" is displayed			
	I.1.2		ST-122	Click the "Upload photo" button and choose a picture	The new profile photo is displayed			
	I.1.3		ST-123	Click the "Remove" button	The profile photo is removed			
	I.2		ST-124	Click the "Password" tab, enter the current, new and confirm password, then click the "Update password" button	"Password changed" is displayed			
	I.3		ST-125	Click the "Preferences" tab, enter the cashier discount limit, then click the "Save limit" button	"Cashiers can now discount up to (n)%" is displayed			
	I.3.1		ST-126	In Appearance, choose a mode, font and text size, then click the "Save changes" button	"Appearance saved" is displayed			
	I.3.2		ST-127	Change the appearance, then click the "Discard" button	"Changes discarded" is displayed and the appearance goes back			
	I.3.3		ST-128	Click the "Reset" button	The default appearance is restored			
	I.4		ST-129	Click the "Session" tab	The full name, role and account status of the signed-in user are displayed			
	I.4.1		ST-130	Click the "Sign out" button	The Logout question is displayed (see L.1)			
	I.5		ST-131	Click the "Email" tab, enter the Gmail address and app password, then click the "Save & Check" button	The settings are saved and checked with Gmail			
	I.5.1		ST-132	Enter a recipient, then click the "Send test email" button	The test email is sent			
J LOGOUT								

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	J.1		ST-133	Click the Logout icon in the sidebar	The question "Are you sure you want to logout?" is displayed			
	J.1.1		ST-134	Click the "Yes, Logout" button	Redirects to the Sign in page			
	J.1.2		ST-135	Click the Logout icon, then the "Cancel" button	The question closes and the administrator stays on the same page			

How to follow this plan

The plan walks every screen the Administrator uses, in the order they meet them: A Sign in, B Dashboard, C Inventory, D Sales, E Suppliers, F Reports, G Notifications, H Users, I Settings and J Logout.

Each screen is numbered by step and sub-step (C.4 opens a window, C.4.1, C.4.2 … are done inside it). Every button is tried, including Cancel, Close, Done and the Cancel of each question, and each expected result says what appears and where the user is taken. Run the steps from top to bottom.

Appendix D

System Test Plan (Cashier)

SYSTEM TEST PLAN — CASHIER		Legend:	Passed (Result is same as expected)
Project Name:	FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies		Failed (Actual result is NOT same as expected result)
Project Manager:	Juan Carlo Reusora and Joylinda Jugan		In Progress (Test is still in progress or not yet done)
Date:			Open (Not yet tested)
Run/Tested by:			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
CASHIER								
A SIGN IN								
	A.1		ST-001	Open FETCH in the browser	The Sign in page is displayed with Username, Password, "Keep me signed in" and "Forgot your password?"			
	A.2		ST-002	Enter the username and the password, then click the "Sign in" button	Redirects to the Point of Sale, the Cashier's home page			
	A.2.1		ST-003	Click the eye icon in the Password field, then click it again	The password is shown as plain text, then hidden again			
	A.2.2		ST-004	Tick "Keep me signed in", then click the "Sign in" button	Redirects to the home page; after the browser is closed and opened again, the user is still signed in			
	A.3		ST-005	Click "Forgot your password?"	The Reset your password window is displayed, with the username already filled in			
	A.3.1		ST-006	Enter the username or email, then click the "Ask for a reset" button	The message "If that account exists: an administrator account is emailed a reset link straight away. Any other account waits for an administrator to approve it, and is then emailed the link." is displayed			
	A.3.2		ST-007	Click the "Cancel" button	Closes the window and returns to the Sign in page			
	A.3.3		ST-008	Open the reset link from the email	The Choose a new password page is displayed			
	A.3.4		ST-009	Enter and confirm the new password, then click the "Save new password" button	"Password has been reset. You can now sign in. Redirecting to sign in..." is displayed, then redirects to the Sign in page			
	A.4		ST-010	Open the confirmation link sent to a new account's email	The page says the address is confirmed and the account can now sign in			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
B POINT OF SALE								
	B.1		ST-011	Sign in as Cashier	The Point of Sale opens with the Starting cash window (first sign-in of the day)			
	B.1.1		ST-012	Enter the cash in the drawer, then click the "Save starting cash" button	"Starting cash saved: ₱(amount)" is displayed			
	B.2		ST-013	Click a category button, then type a product name in the search box	Only the matching products are shown			
	B.3		ST-014	Click a product card	"(product) added to cart" is displayed and it is in the Current sale			
	B.3.1		ST-015	Click the + and − buttons of the item	The quantity and the total change			
	B.3.2		ST-016	Click the "Clear" button	The cart is emptied			
	B.4		ST-017	Add an item, then click the quick cash buttons ("Exact", "+20", "+50", "+100", "+500", "+1000")	The cash received is filled in and the change is worked out			
	B.5		ST-018	Click the "Charge" button	The Confirm cash sale window is displayed			
	B.5.1		ST-019	Click the "Complete sale" button	"Sale completed! Change: ₱(amount)" is displayed, the stock goes down and the receipt prints			
	B.5.2		ST-020	Click "Charge", then the "Cancel" button	The window closes and the cart stays as it was			
	B.6		ST-021	Click the "My sales today" button	This cashier's sales of today are listed			
	B.6.1		ST-022	Click "Reprint" on a sale	The receipt prints again			
	B.6.2		ST-023	Click "Ask for void" on a sale, enter what went wrong, then click "Send request"	"Sent to a manager" is displayed			
	B.6.3		ST-024	Click the "Close" button	Closes the window and returns to the Point of Sale			
	B.7		ST-025	Click the "Cash in / out" button	The cash in and out window is displayed with today's movements			
	B.7.1		ST-026	Choose "Taken out", enter the amount and the reason, click "Record cash taken out", then "Record it"	"Recorded" is displayed and the movement is listed			
	B.7.2		ST-027	Click the "Close" button	Closes the window and returns to the Point of Sale			
	B.8		ST-028	Click the "End of day" button	The End of day window is displayed with the starting cash, cash sales and cash in and out; the expected cash is not shown until the count is saved (a blind count)			
	B.8.1		ST-029	Click the "Change" button beside the starting cash	The Starting cash window opens with a "Cancel" button			
	B.8.2		ST-030	Enter the cash counted, then click the "Save count and close drawer" button	The count is saved, the expected cash is shown with Balanced, Over by or Short by, and the drawer is closed for the day			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	B.8.3		ST-031	Click "End of day", then the "Close" button	Closes the window and returns to the Point of Sale			
	B.9		ST-032	Click the Settings (gear) icon	The Profile tab is displayed with the cashier's own name and email			
	B.9.1		ST-033	Click the "Password" tab, enter the current, new and confirm password, then click the "Update password" button	"Password changed" is displayed			
	B.9.2		ST-034	Click the "Session" tab	The full name, the role "Staff" and the account status are displayed			
	B.9.3		ST-035	Click "POS" in the sidebar	Returns to the Point of Sale			
C LOGOUT								
	C.1		ST-036	Click the Logout icon in the sidebar	The question "Are you sure you want to logout?" is displayed			
	C.1.1		ST-037	Click the "Yes, Logout" button	Redirects to the Sign in page			
	C.1.2		ST-038	Click the Logout icon, then the "Cancel" button	The question closes and the cashier stays on the same page			

How to follow this plan

The plan walks every screen the Cashier uses, in the order they meet them: A Sign in, B Point of Sale (with the cashier's Settings) and C Logout.

Each screen is numbered by step and sub-step (C.4 opens a window, C.4.1, C.4.2 … are done inside it). Every button is tried, including Cancel, Close, Done and the Cancel of each question, and each expected result says what appears and where the user is taken. Run the steps from top to bottom.

Appendix E

System Test Plan (Supplier)

SYSTEM TEST PLAN — SUPPLIER		**Legend:**	**Passed (Result is same as expected)**
Project Name:	FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies		**Failed (Actual result is NOT same as expected result)**
Project Manager:	Juan Carlo Reusora and Joylinda Jugan		**In Progress (Test is still in progress or not yet done)**
Date:			**Open (Not yet tested)**
Run/Tested by:			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
SUPPLIER								
A SIGN IN								
	A.1		ST-001	Open FETCH in the browser	The Sign in page is displayed with Username, Password, "Keep me signed in" and "Forgot your password?"			
	A.2		ST-002	Enter the username and the password, then click the "Sign in" button	Redirects to the Supplier Portal, the Supplier's home page			
	A.2.1		ST-003	Click the eye icon in the Password field, then click it again	The password is shown as plain text, then hidden again			
	A.2.2		ST-004	Tick "Keep me signed in", then click the "Sign in" button	Redirects to the home page; after the browser is closed and opened again, the user is still signed in			
	A.3		ST-005	Click "Forgot your password?"	The Reset your password window is displayed, with the username already filled in			
	A.3.1		ST-006	Enter the username or email, then click the "Ask for a reset" button	The message "If that account exists: an administrator account is emailed a reset link straight away. Any other account waits for an administrator to approve it, and is then emailed the link." is displayed			
	A.3.2		ST-007	Click the "Cancel" button	Closes the window and returns to the Sign in page			
	A.3.3		ST-008	Open the reset link from the email	The Choose a new password page is displayed			
	A.3.4		ST-009	Enter and confirm the new password, then click the "Save new password" button	"Password has been reset. You can now sign in. Redirecting to sign in..." is displayed, then redirects to the Sign in page			
	A.4		ST-010	Open the confirmation link sent to a new account's email	The page says the address is confirmed and the account can now sign in			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
B SUPPLIER PORTAL								
	B.1		ST-011	Sign in as Supplier	The Supplier Portal is displayed with the figures Orders waiting for you, On the way, Waiting to be paid and Running low at the shop			
	B.2		ST-012	Look at Open orders	Only this supplier's waiting, confirmed and shipped orders are listed			
	B.3		ST-013	Click "Confirm order" on a waiting order, choose the delivery date, then click the "Confirm order" button	"Order confirmed." is displayed			
	B.3.1		ST-014	Click "Confirm order", then the "Cancel" button	Closes the window; the order is still waiting			
	B.4		ST-015	Click "Mark as shipped", enter the delivery receipt number, then click the "Mark as shipped" button	"Order marked as shipped." is displayed			
	B.4.1		ST-016	Click "Mark as shipped", then the "Cancel" button	Closes the window; the order is still confirmed			
	B.5		ST-017	Click "Change delivery date", choose the new date, enter the reason, then click "Save new date"	"Delivery date updated." is displayed			
	B.5.1		ST-018	Click "Change delivery date", then the × (Close) button	Closes the window; the delivery date stays the same			
	B.6		ST-019	Click "Notes" on an order, type a note, then click "Send"	The note is added for the shop to read			
	B.7		ST-020	Click the "Offer an order" button, enter the quantities and the date, then click "Send offer"	"Offer sent. It becomes an order if the shop accepts it." is displayed			
	B.7.1		ST-021	Click "Offer an order", then the "Cancel" button	Closes the window; no offer is sent			
	B.8		ST-022	Look at Payments	The delivered orders the shop has not yet marked as paid are listed			
	B.9		ST-023	Look at Products you supply	Only this supplier's products are listed, with the stock at the shop, reorder level, status, what the shop expects to sell in the next 30 days and your price			
	B.10		ST-024	Click "Change price" on a product, enter the new price, then click "Send to the shop"	"Sent to the shop for approval. The current price stays until they agree." is displayed			
	B.10.1		ST-025	Click "Change price", then the "Cancel" button	Closes the window; the price stays the same			
	B.11		ST-026	Look at How you are doing	The star rating out of 5 is displayed, with whether it meets the standard (4 stars or more) and the figure for each condition			
	B.12		ST-027	Look at Past orders	This supplier's received and cancelled orders are listed			
	B.13		ST-028	Click the "Edit contact details" button, change them, then click "Save details"	"Your details were updated." is displayed			
	B.13.1		ST-029	Click "Edit contact details", then the "Cancel" button	Closes the window; the details stay the same			

MODULE	NO.	TEST CASE	ID	DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS	COMMENT
	B.14		ST-030	Click the Settings (gear) icon	The Profile tab is displayed with the supplier account's own name and email			
	B.14.1		ST-031	Click "Orders and products" in the sidebar	Returns to the Supplier Portal			
C LOGOUT								
	C.1		ST-032	Click the Logout icon in the sidebar	The question "Are you sure you want to logout?" is displayed			
	C.1.1		ST-033	Click the "Yes, Logout" button	Redirects to the Sign in page			
	C.1.2		ST-034	Click the Logout icon, then the "Cancel" button	The question closes and the supplier stays on the same page			

How to follow this plan

The plan walks every screen the Supplier uses, in the order they meet them: A Sign in, B Supplier Portal (with the supplier's Settings) and C Logout.

Each screen is numbered by step and sub-step (C.4 opens a window, C.4.1, C.4.2 … are done inside it). Every button is tried, including Cancel, Close, Done and the Cancel of each question, and each expected result says what appears and where the user is taken. Run the steps from top to bottom.